

This google sheet is created by Dr Mostafa Saad Ibrahim . Overall ~950 problems for newcomers to problem solving.					
Problem Solving Sheet	mostafa.saad.fci@gmail.com		Ask.fm	Site / More Contacts	
	Video Introducing roadmap (Arabic) - to min 18 ONLY			Video explaining the sheet	
	Currentnet Version V7.0		Latest Version		
	- Complete and consistent roadmap for newcomers: What to solve & algorithms to learn in order - In the bottom row, there are different sheet pages such as Faq, Topics, CF-C2 - CF-C1, C2 are (Codeforces Div2 C problems (or similar level from other OJs), but from easy to hard). Same for CF-D1, D2, D3 - Covering most of topics needed up to codeforces Div2-D - Problems of scales 1 - 5.5 / 10 + Few harder ones - Problems increase in difficulty per topic with intermediate easy/medium problems + ad-hoc problems - Speed problems to maintain speed goals - A lot of recorded videos for problems solutions, especially for the entry levels (Arabic) - Several students followed its order and managed to solve by themselves 95% of it (up to his current sheet page) - You can train in one of the following ways: - A) Blind-Order training style - Problems are distributed in sheets CF-A, CF-B, CF-C1,CF-D3 - This one is a roadmap. It targets learning the knowledge/skills in a consistent and balanced way - Every sheet page is on average harder than the previous sheet page - This is my recommended way, though most camps/training-approaches don't use this style - B) Topics-Based training style - See sheet page (Topics1). It has the same sheet problems (CF-A to CF-D3) ordered by category and level, around 950 problems - Ideas Quality column: P5 (important), P4(very interesting), P3(interesting), P2(good), P1(ok), Empty (normal) - Say your level is 6/10, and solved a problem of level 3 with P5, you will find it a normal one. So notice, it is subjective to your level/background - You can train using Blind-Order, and use Topics page as guide to skip some problems - Many guys/training camps are fan of this topics-based way. - You need to be careful with such style as it may corrupt your training quality, e.g. due to your bias - Advantage: Mastering the algorithm till solving some hard problems in a short time - Disadvantage: Discovering the algorithm behind the problem is an important skill. Given that you know the topic, you lose a good space to improve this skill - Disadvantage: Being in the mode of specific algorithm lets you solve many of it easier. However, when solving in real contests, your mind is not so active on the specific topic - It is still a good training roadmap. Actually used by most of people I think. - See Topics2 page (for extra topics/problems in specific cases)				
What is this Sheet?					
Advantages of this Sheet?	- To be a strong contestant, one has to take care of a number of quality and quantity factors ==> This roadmap does its best to satisfy that - Typical issues in our Arabian region: Guys with 700-1000 solved problems and still weak! - Why? A) No specific roadmap or keep switching between them B) Training while knowing problem category / level C) Focus on specific online judge - Again, this sheet solves these issues - Allows you to write down your statistics to learn from them (e.g. you consume much time in debugging) - Continuous refining based on feedback				
Your Sheet COPY	This is a personal Google sheet for you [Make a copy from file MENU] to have sets of problems to solve coupled with algorithms to learn - Don't download the sheet, Use it online - Can't edit it? Because it is read-only. Read below notes. - Just make a copy to your google driver - Then work over it online. Following are the details - Login to ur google Gmail - Go to my sheet - In the sheet click on 'file' menu - select Make copy - it will create copy for u - RENAME it to Junior Sheet <Your name> - Now the copied sheet is opened for you (or go inside ur Google drive and you will find it) NOTE: If u did so and still read-only format, then you are again opening my sheet (e.g. with old name), NOT your copy				
For Whom?	A junior is anyone who doesn't master solving codeforces Div2-D.				
Prerequisites?	Basic Programming skills such in series C++ Programming + STL + Debugging Skills				C++ is highly recommended
	If you find my sheet is hard, Finish Assiut University provides an easier starting roadmap. Finish it first				Novice RoadMap Online Judge
	Know about our community and what is programming competitions ==>				Watch these videos for more details
	Code with any language but preferred C++ or Java. For Java : Solver to be Channel				Code El Masry Channel
Training Style?	You can train alone, but highly advised to find partner(s) to work with to encourage each other.				
Skills Goals	Moving from Junior Level to Semi-Senior Level: A one who do pretty well in CF-Div2 A, B, C, D and similar levels (e.g. TC-Div2-1000)				
Knowledge Goals	Understand and build fair knowledge in some algorithms in Number Theory, Dynamic Programming, Greedy, Graph Theory and Search				
Sheets	Sheet pages are mainly for Codeforces Div2 A, B, C, D + Problems on knowldge topics (Mainly from UVA, SPOJ)				
	Each sheet has some sets, each set is ~10-15 problems....The top sets are mandatory....The below sets (after line mark) are optional				
	If you did well in the mandatory sets, move to next sheet...otherwise you still need training on similar level...then solve the optional problems				
	Please watch the videos in order, solve UVA/SPOJ problems in order. Don't skip them.				
	In some columns, some time recordings. This helps you to know how much time you take per a problem...use that to recognize your problems				
	In the level column give an estimate to the problem level from 1-2 (easy), 3-4 (medium), 5-6(hard), 7-8 (had to read editorials), 9-10 (can't solve)				
	In the comments column..write comments for hard problems.				
	Put problem Status AC (for Accepted) CS (can't solve) Other values: WA (wrong answer), TLE (time limit exceeded), RTE, MLE				
	If you solved a problem before, put ACX instead of AC. Don't resolve				
	Don't let a problem consumes more than 2-3 hours. If can't solve it, see editorials/solutions. If still can't solve it, just leave it for now.				
	Don't compare yourself with others. People vary in their progress				

Thanks for all guys who sent sheet feedback: Mariam Alshereef, Magdy Hassan, Ahmed Yasser, Ahmed Elsayed Awad, Mohamed Nasser, Mostafa Ali Mansour, Aya elymany, Ayyad shenouda, Others.
Special Thanks for Coach Alhussain Aly for his continuous help
Special Thanks for All volunteers in videos recording / Editorials writing

Q) What is the sheet requirements? Should I study algorithms and Data structures?

- ONLY programming skills (e.g. Programming 1 level). It is highly advised to implement 2-3 projects
- NO for OOP
- NO for datastructures, but learn STL (or Collections in Java/C#). It helps alot
- NO for algorithms, the sheet will teach you that in a smooth way
- For C++ guys (and others as guide) - first 18 videos here: <https://www.youtube.com/playlist?list=PLPt2dINI2MIZPFq6HyUB1Uhxh1UDnZMS>

Q) How much time do I need to finish the sheet?

- Answer varies from one to another.
- Some trainees are fast through whole sheet
- Some trainees are slow through whole sheet
- Some trainees are fast in early pages, but slow at the end
- Some trainees are slow in early pages, but fast at the end
- The sheet has ~900 problems. Around 60 videos. Problems targets average guys. However, you are encouraged to skip problems whenever you could. I expect many guys could skip 20%-30% of the mandatory problems. Make use of the Topics page.

215 problems of level <= 2.5 (avg 20 min per problem)
93 problems of level <= 3.5 (avg 30 min per problem)
270 problems of level <= 4.5 (avg 40 min per problem)
178 problems of level <= 5.25 (avg 60 min per problem)
127 problems of level <= 5.75 (avg 75 min per problem)
53 problems of level > 5.75 (avg 90 min per problem)

$215 \times 20 + 93 \times 30 + 270 \times 40 + 178 \times 60 + 127 \times 75 + 53 \times 90 = \sim 700$ hours (say max 900 hours)

If you trained in the summer vacation seriously for 2 month (e.g. 10 hours * 30 days * 2 month = 600 hours) + the remaining of the year effort, you could solve the whole sheet smoothly and move to Div2-E level goal

- <https://ask.fm/mostafasaad87/answers/144907000290> [adjust to whatever fits with you]

Q) When should I give up and check the editorials and solutions?

<https://ask.fm/mostafasaad87/answers/144907000290>

Q) Got WA, should I check directly the test cases?

- No, remember in a real contest you only know your problem status (WA, TLE, ...etc)
- Struggle to find the wrong case by yourself. At least 15-30 minutes.
- Don't keep trying longer, just check the test cases
- If you can write a brute force solution for your problem, write a stress test: Generate random cases and compare the optimal algorithm with the brute force case

Q) What is the debug time?

- Once you finish coding and start testing, you verify if the program is working as expected or not
- If not, there are bugs that you need to find to make the program behave as expected. From this moment till getting the program AC = debugging time
- People could debug using 'print statements'. A better way using a debugger
- Check out these 4 videos: <https://www.youtube.com/watch?v=DlbQwQEiDW0&t=0s&index=35&list=PLPt2dINI2MIZPFq6HyUB1Uhxh1UDnZMS>

Q) Should I solve every problem?

- Generally, preferred, but If you think certain level is easy (e.g. solve it within 15 minutes), then jump a block and so on
- This jumping might be for codeforces problems only

Q) Just started in Div2-A, could I finish its codeforces problems first, then solve the UVA/Colored problems?

- Many juniors find UVA problems in Div2-A hard, I understand
- Yes, almost same for Div2-B. But don't do that in next sheets as order might matter, because all of such knowledge are mainly preparations for hard Div2-B or Div2-C
- However, following the order is a much better idea
- Similarly, one could finish All Div2-A/Div2-B codeforces problems, then solve their colored problems. Again, this is not the best way.

Q) Is using C# ok?

- Generally yes, but you won't be able to submit in UVA judge, as C# is not supported
- For such problems, write your code, but heavily test it. You may download an internet code and evaluate the test case on both
- On the other hand, learning Basic C++ + STL is not hard for C#/Java guys
- C++/Java/Python are official in UVA
- Codeforces allows more such as Javascript

Q) When I watch a video, should I solve the problems in its info section?

- No, sheet has subset of these problems already in specific order
- Sheet is self-contained

Q) I watched the video, but it is hard, any tips?

- Algorithms are hard, learn to struggle
- Watch the video 2-3 times, try to rewrite its code by yourself
- Still can't get it? Google for more materials from the web (ppt/pdf/videos) and try to learn
- In worst case, leave it for now and return to it later

Q) How does your sheet prepare for ECPC/ACPC?

- The sheet prepares you to reach level 5-5.5/10 in several categories
- If a team of 3 members solved the whole sheet, they may rank in the top 15 in the contest

But let's go in details. Individual success in contests depends on several factors. Let me state some of them.

- Solving many problems of good quality
- Improving your different skills (reading, thinking, coding and accuracy).
- 2 persons could solve in training the same problem. One got it in 20 min first submission, and the 2nd needed 90 minutes due to 60 minutes debugging.
- Healthy training: Regular / good times for training (e.g. morning) / weekly contests / reading other codes / collaboration with others / etc
- Stress management during contests
- Emotions management when fail in solving or feel performance is not improving enough
- Avoiding Psychological issues: Comparing to others, Negative feelings, Your image, Regretting training time

Moving toward a team contest, you need more concerns:

- Serious team members. If only one active member, they may end up in bad performance. So EACH team member need to finish the sheet individually + weekly contests
- Tolerating team mistakes during the contest
- PC management
- Suitable strategy + several team contests to tune it

As you may notice, there are MANY factors for success.

- This sheet provides you with high quality problems and good topics distribution + way to record your stats to know your weak points
- However, there are many concerns that YOU have to tackle by yourself and your team members

- Finish up to CF-C2 sheet, then study from the "Cracking the Coding Interview: 150 Programming Questions and Solutions" book
- Also watch: <https://www.youtube.com/watch?v=39vqarATPyM>

Q) How different is your sheet versus Ahmed Aly Ladders?

- Ladder problems are selected automatically, no personal investigation for the actual benefit/need from the problem
- Mine is mixed between automated and manual.
- At the current moments, many of my trainees and students feedback, I am aware of the problem level and its category.
- I updated the sheet many times because of the received feedback
- My sheet involves the algorithms videos to learn, in order, while you grow up.
- I selected videos to prepare you as soon as possible for Div2-C/Div2-D where many algorithms starts to appear
- It is a sheet..ready for you to record your times, notes...etc...this help to improve yourself
- It is not blocking style. If you can't solve problem, just leave it and move to other one. In ladders, you see next problem when solve current one (or do workarounds)

Q) How did you select problems for the sheet?

- Long story, many versions were there, from a version to another improvements were applied
- Codeforces problems where rated based on this CF tool: http://codeforces.com/blog/entry/46304?mobile=true#_=_
- Any rating is just an estimation. I found this one a pretty reasonable measure
- The videos are selected such that when comes to Div2-C/, you are ready
- Manual selections and investigations for non-CF problems to be used in the sheet
- Lots of manual efforts and investigations and feedback processing

Q) what is the next step after finishing your sheet ?

- Joining directly my ICPC semi-seniors supervision, **BUT**
- Email me with your online sheet copy link and it **must** have
- Each row should have: code link, time details, problem level, category and comment per a problem
- I will review and decide
- Side note: If you started in Div2-C1 and solved first 15 problems, you can share the sheet with me to follow your updates

Q) can't access the sheet in edit mode?

- Don't download the sheet, Work over it online "better"
- Can't edit it? Because it is read-only. Read below notes.
- Just make a copy to your google driver
- Then work over it online. Following are the details

- Login to ur google Gmail
- Go to my sheet
- In the sheet click on 'file' menu
- select Make copy
- it will create copy for u
- RENAME it to Junior Training Sheet
- Now the copied sheet is opened for you (or go inside ur Google drive and you will find it)

NOTE: If u did so and still read-only format, then you are again opening my sheet (e.g. with old name), NOT your copy

Q) What to write in the category column?

The algorithm used to solve a problem. In Div2-A, this might be:

- This column is for the algorithm you used during solving. Usually, new guys in CF-A are confused. If so, leave it CF-A and start to write in CF-B
- The more you go in the sheet you will learn algorithms (e.g. Binary search, DP, DFS, etc). Then this what you write in level column
- The problem that has no algorithm but a specific idea called ad-hoc, This is the case for most of CF-A and less later
- Implementation: Means the problem request is almost direct, just code it
- Brute Force: Means instead of finding elegant solution, try all possible solutions (e.g. 3 nested loops) and select the solution
- Ad-hoc: Just per-problem thinking in a special way/analysis on how to solve the problem
- Please watch from this minute: <https://youtu.be/DZ6YtILCE8?t=839>

Q) Are problems really sorted based on easiness? I don't feel so.

They are sorted by easiness already. But, whatever order, anyone will find some are easy and some are hard in some order.

That is, no one can give you a list that every problem for YOU is easier than the next problem.

In other words, If we gave 100 problems to 10 students of same level to solve and told them rank from easy to hard, they will rank them differently.

So, questions ordered by people average. The promise is, the problems will be within your range to solve.

Q) What are these problems colors?

See "Problems Colors" notes in info page

Q) Are the problems sorted?

Yes, but this is tricky as sorting is subjective.
That is imagine 10 prblms given for 100 people to order based on its level, people will arrange in different ways based on the experience

So if you felt they are not sorted, just keep going

Q) Why problem-solving is that important?

See the first 2 videos here: <https://www.youtube.com/playlist?list=PLPt2dINI2MIaNCuJ070HIAO8JWYBcafuyG>

Q) I feel bored when solving problems compare to doing projects?

<https://ask.fm/mostafasaad87/answers/145333554402>

Q) I would like to freeze my study for 1-2 years to be good in problem-solving?

I never liked that. Graduate on time. In your free times and vacations do more problem solving
Relevant: <https://ask.fm/mostafasaad87/answers/145151822818>

Q) Topics based-training vs Blind Order

In topics training, we study a topic, then solve a lot of problems over it.

Advantages:

- Mastering the algorithm till solving some hard problems in short time

Disadvantages:

- Discovering the algorithm behind the problem is an important skill. Given that you know the topic, you lose a good space to improve this skill
- Being in the mode of specific algorithm lets you solve many of it easier. However, when solving in real contests, your mind is not so active on specific topic

In my sheets - Blind style:

- You solve 3-5 per topic. Then you have to discover the other problems by yourself. So you train to avoid the missing 2 points

Claim:

- Although topics training let guys be so good early, they level stuck early and they don't improve. Seems to me, topics training is an important factor in doing so. Meanwhile, if you just target to be good in Div2D level in shorter time and no interest in further competitions achievements, you may go topics based.

Q) Who Finished my sheet? Their levels?

<https://ask.fm/mostafasaad87/answers/150802497762>

Q) How to share my sheet progress with you?

<https://ask.fm/mostafasaad87/answers/148552940002>

Q) What is after the sheet?

- There are 2 other levels, each has around 1000 problems. Semi-senior level and seniors level
- Generally speaking, the region stars will solve a lot of problems, e.g. 2000-3000 problems with many of them of hard level
- Whoever finish the sheet, I join him in my supervision for the next levels

Problem Name	Problem Code	Status	Submit Count	Reading Time(m)	Thinking Time(m)	Coding Time(m)	Debug Time(m)	Total Time(m)	Problem Level /10	By yourself?	Category	1-2 line Comments About your approach
	AC Averages ==>	3	2.3	5	13	15	18	50	2	2	2	2
Sample Name1	Sample Link1	AC	5	4	8	6	32	50	2	Yes	Math	Solution in mind is O(n^2): for each pair of points, get the equation of the straight line linking between them, and add 2 to its count. Print the count of the line having the max count. Problem is how to hash a line equation (coefficients are double).
Sample Name2	Sample Link2	AC	1	5	10	35	20	70	2	No	Impl	My Performance Notes: This is so bad performance. Needed many submissions per problem. Always submit as if you are in real contest. Submit to AC, not to see if we will pass or not. Target AC from 1st submission. Think more before submission.
Sample Name3	Sample Link3	AC	1	5	20	4	1	30	2	Yes	Graph	Please always write and study your timings.
Sample Name4	Sample Link4	WA	5	4	25	20	2	51	7	Hint	Math	My Performance Notes: This is so good performance. 1st submission. Thinking is the higher. Code/Debug is so low. By time, thinking column will be improved.
Sample Name5	Sample Link5	CS	6	5	30	25	31	91	9			Other Status values: AC, WA, CS, TLE, MLE, RTE, ... These values and comments are just examples. Just remove/ignore them.
								0				Want c++ solution for UVA 408? Google with: UVA 408 filetype:c++
								0				Watch - Approaching Problem Statement
Vanya and Feni	CF677-D2-A							0				Watch - Thinking - On papers Not on PC
Anton and Dani	CF734-D2-A							0				C++ Solution Example
								0				This is from Round 379. Here is the editorial
								0				<i>You shouldn't watch a solution video unless you can't solve it by yourself and don't get it from editorial/code. Videos are there just to for extra help.</i>
								0				In the first 20 problems, don't think more than 20 minutes. After that see the solutions.
								0				
Bear and Big Bl	CF791-D2-A							0				Video Solution - Eng Youssef El Ghareeb
Team	CF231-D2-A							0				Video Solution - Eng Youssef Ali
Beautiful Matrix	CF263-D2-A							0				Video Solution - Eng Samed Hajaj
Gravity Flip	CF405-D2-A							0				Video Solution - Eng John Gama
Petya and Strin	CF112-D2-A							0				Video Solution - Solver to be (Java)
Boy or Girl	CF236-D2-A							0				Video Solution - Solver to be (Java)
Word	CF59-D2-A							0				Video Solution - Solver to be (Java)
Magnets	CF344-D2-A							0				Video Solution - Solver to be (Java)
Sereja and Dim	CF381-D2-A							0				Video Solution - Solver to be (Java)
Stones on the T	CF266-D2-A							0				Video Solution - Eng Ahmead Raafat (Python)
Police Recruits	CF427-D2-A							0				Video Solution - Eng Ahmead Raafat (Python)
Black Square	CF431-D2-A							0				Video Solution - Eng Ahmead Raafat (Python)
Night at the Mu	CF731-D2-A							0				Video Solution - Eng Yahia Ashraf
Games	CF268-D2-A							0				Video Solution - Eng Yahia Ashraf
								0				
								0				Watch - Measuring Algorithms Performance - 1
								0				Watch - Elementary Math - Introduction
Buy a Shovel	CF732-D2-A							0				Video Solution - Eng Yahia Ashraf
Is your horse sh	CF228-D2-A							0				Video Solution - Eng Ahmead Raafat (Python)
Colorful Stones	CF265-D2-A							0				Video Solution - Eng Ahmead Raafat (Python)
Die Roll	CF9-D2-A							0				Video Solution - Eng Muntaser Abukadeja
Shaass and Osi	CF294-D2-A							0				Video Solution - Dr Mostafa Saad
Juicer	CF709-D2-A							0				Video Solution - Solver to be (Java)
Carrot Cakes	CF799-D2-A							0				Video Solution - Solver to be (Java)
Anton and Letti	CF443-D2-A							0				Video Solution - Solver to be (Java)
Way Too Long I	CF71-D2-A							0				Video Solution - Solver to be (Java)
Free Ice Cream	CF688-D2-A							0				Video Solution - Solver to be (Java)
Helpful Maths	CF339-D2-A							0				Video Solution - Solver to be (Java)
Team Olympiad	CF490-D2-A							0				Video Solution - Eng Muntaser Abukadeja
New Password	CF770-D2-A							0				
								0				
								0				Watch - Number Theory - Modular Arithmetic
								0				Watch - Combinatorics - Counting Principles
Light, more ligh	UVA 10110							0				Video Solution - Eng Amr Saud
Product	UVA 10106							0				Video Solution - Eng Youssef El Ghareeb, Don't solve using big integer
Uniform Genera	UVA 408							0				Video Solution - Eng Yahia Ashraf
Black and white	UVA 11231							0				Video Solution - Eng Amr Saud
	SPOJ EASYMATI							0				Sol
Electricity	UVA 12148							0				Learn Calender Leap Year
								0				
Presents	CF136-D2-A							0				Video Solution - Eng Ahmed Rafaat (Python)
Lineland Mail	CF567-D2-A							0				Video Solution - Eng Ahmed Rafaat (Python)
Mahmoud and L	CF766-D2-A							0				Video Solution - Solver to be (Java)
Snacktower	CF767-D2-A							0				
Oath of the Nigl	CF768-D2-A							0				Video Solution - Solver to be (Java)
Pangram	CF520-D2-A							0				Video Solution - Solver to be (Java)
Twins	CF160-D2-A							0				Video Solution - Solver to be (Java)
Keyboard	CF474-D2-A							0				Video Solution - Solver to be (Java)
								0				
								0				Watch - Graph Theory - Intro
								0				Watch - Graph Theory - DFS
The Seasonal V	UVA 352							0				Video Solution - Eng Mohamed Nasser
Marcus	UVA 10452							0				Video Solution - Eng Ayman Salah
Battleships	UVA 11953							0				Video Solution - Eng Aya Elymany
								0				Read definition of: Bipartite graph
Forming Teams	CF216-D2-B							0				Video Solution - Dr Mostafa Saad
Hierarchy	SPOJ MAKETREI							0				Video Solution - Eng Yahia Ashraf
Ordering Tasks	UVA 10305							0				Video Solution - Eng Yahia Ashraf
								0				
Even Odds	CF318-D2-A							0				Video Solution - Eng Muntaser Abukadeja
I Wanna Be the	CF469-D2-A							0				Video Solution - Solver to be (Java)
Is it rated?	CF807-D2-A							0				Video Solution - Solver to be (Java)
Olesya and Roc	CF584-D2-A							0				Video Solution - Solver to be (Java)
Football	CF43-D2-A							0				Video Solution - Eng Belal Abdunnasser (Python)
Brain's Photos	CF707-D2-A							0				Video Solution - Solver to be (Java)
Dubstep	CF208-D2-A							0				Video Solution - Solver to be (Java)
Valera and X	CF404-D2-A							0				Video Solution - Solver to be (Java)
Arpa's hard exa	CF742-D2-A							0				Video Solution - Solver to be (Java)
Calculating Fun	CF486-D2-A							0				Video Solution - Solver to be (Java)
Theatre Square	CF1-D12-A							0				Video Solution - Solver to be (Java)
Anton and Poly	CF785-D2-A							0				Video Solution - Solver to be (Java)
Panoramix's Pri	CF80-D2-A							0				Video Solution - Solver to be (Java)

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	AC Averages =>	3	2.3	5	13	15	18	50	2	2	2	2
Counterexample	CF483-D2-A							0				Video Solution - Solver to be (Java)
								0				
								0				Watch - Computational Geometry - Intro
								0				Watch - Computational Geometry - Point and Vector
Wasted Time	CF127-D2-A							0				
Points in Figure	UVA 476							0				
Overlapping Rectangles	UVA 460							0				Video Solution - Eng Muntaser Abukadeja
Fancy Fence	CF270-D2-A							0				Video Solution - Eng Omar Ashraf
Pouring Rain	CF667-D2-A							0				
Fourth Point !!	UVA 10242							0				Video Solution - Eng Magdy Hasan
								0				
Good Number	CF365-D2-A							0				Video Solution - Eng Muntaser Abukadeja
Dice Tower	CF225-D2-A							0				Video Solution - Eng Muntaser Abukadeja
Alyona and Nuriya	CF682-D2-A							0				Video Solution - Eng John Gamal
Mountain Scenery	CF218-D2-A							0				Video Solution - Eng John Gamal
Help Vasilisa the Princess	CF143-D2-A							0				Video Solution - Eng John Gamal
Chewbacca and Han	CF514-D2-A							0				Video Solution - Eng Muntaser Abukadeja
Ksenia and Pan	CF382-D2-A							0				Video Solution - Eng Samed Hajajla
Launch of Collets	CF699-D2-A							0				Video Solution - Eng Samed Hajajla
Polo the Penguin	CF289-D2-A							0				Video Solution - Dr Mostafa Saad
IQ Test	CF267-D2-A							0				Video Solution - Dr Mostafa Saad
Yaroslav and Perpetua	CF296-D2-A							0				Video Solution - Dr Mostafa Saad
Snow Footprints	CF298-D2-A							0				Video Solution - Dr Mostafa Saad
Raising Bacteria	CF579-D2-A							0				Video Solution - Eng Ahmed Rafaat (Python)
BowWow and the Fire	CF1204-D2-A							0				Video Solution - Dr Mostafa Saad
Balanced Rating	CF1237-D12-A							0				
								0				
								0				Watch - Search Techniques - Binary Search
The Playboy Chimp	UVA 10611							0				Video Solution - Eng Ayman Salah
Pipeline	CF287-D2-B							0				Video Solution - Dr Mostafa Saad
Burning Midnight	CF165-D2-B							0				
Aggressive cow	SPOJ AGGRCOW							0				Video Solution - Eng Youssef El Ghareeb
								0				Before moving to another sheet, email me with feedback about these problems selection.
								0				
								0				You don't have to or encouraged to solve the next problem. If you felt you need so, try some of them. Or Proceed to next and solve in parallel, up to you.
								0				
Word Capitalization	CF281-D2-A							0				Video Solution - Solver to be (Java)
Next Round	CF158-D12-A							0				Video Solution - Solver to be (Java)
Young Physicist	CF69-D2-A							0				Video Solution - Solver to be (Java)
Bit++	CF282-D2-A							0				Video Solution - Solver to be (Java)
Case of the Zeros and Ones	CF556-D2-A							0				Video Solution - Solver to be (Java)
Translation	CF41-D2-A							0				Video Solution - Solver to be (Java)
String Task	CF118-D2-A							0				Video Solution - Solver to be (Java)
Laptops	CF456-D2-A							0				Video Solution - Solver to be (Java)
Left-handers, Right-handers and Computers	CF350-D2-A							0				Video Solution - Eng Hossam Yehia
George and Alex	CF467-D2-A							0				Video Solution - Eng Ahmed Rafaat (Python)
Vasya the Hipster	CF581-D2-A							0				
Fox And Snake	CF510-D2-A							0				
The New Year: A Journey Without Tears	CF723-D2-A							0				
Elephant	CF617-D2-A							0				
Greg's Workout	CF255-D2-A							0				
Ultra-Fast Math	CF61-D2-A							0				
Little Pony and Big Apple	CF454-D2-A							0				
One-dimensional	CF721-D2-A							0				
Soldier and Bananas	CF546-D2-A							0				
								0				
Bus to Udayland	CF711-D2-A							0				
Cookies	CF129-D2-A							0				
Second Order Statistics	CF22-D2-A							0				
Nearly Lucky Numbers	CF110-D2-A							0				
Playing with Dice	CF378-D2-A							0				
A Good Contest	CF681-D2-A							0				
Beautiful Year	CF271-D2-A							0				
Far Relative's Birthday	CF629-D2-A							0				
Mashmokh and Stones	CF415-D2-A							0				
Triangular numbers	CF47-D2-A							0				
								0				
Roma and Luck	CF262-D2-A							0				
Toy Army	CF84-D2-A							0				
Levko and Table	CF361-D2-A							0				
Cards	CF701-D2-A							0				
Wizards' Duel	CF591-D2-A							0				
Combination Lock	CF540-D2-A							0				
Summer Camp	CF672-D2-A							0				
Soft Drinking	CF151-D2-A							0				
Coder	CF384-D2-A							0				
GukiZ and Contests	CF551-D2-A							0				
								0				
Circle Line	CF278-D2-A							0				
Patrick and Sho	CF599-D2-A							0				
Choosing Team	CF432-D2-A							0				
Vanya and Cubes	CF492-D2-A							0				
Insomnia cure	CF148-D2-A							0				
Cakeminator	CF330-D2-A							0				
Flag	CF16-D2-A							0				
Cupboards	CF248-D2-A							0				
Soroban	CF363-D2-A							0				
								0				
Amusing Joke	CF141-D2-A							0				
Lights Out	CF275-D2-A							0				
Lunch Rush	CF276-D2-A							0				
Duff and Meat	CF588-D2-A							0				
Vanya and Carrots	CF401-D2-A							0				
Squats	CF424-D2-A							0				
Arrival of the General	CF144-D2-A							0				
Sinking Ship	CF63-D2-A							0				

Problem Name	Problem Code	Status	Submit Count	Reading Time(m)	Thinking Time(m)	Coding Time(m)	Debug Time(m)	Total Time(m)	Problem Level /10	By yourself?	Category	1-2 line Comments About your approach
	AC Averages =>	3	2.3	5	13	15	18	50	2	2	2	2
LLPS	CF202-D2-A							0				
Candy Bags	CF334-D2-A							0				
								0				
Game With Slic	CF451-D2-A							0				
Vasya and Soci	CF460-D2-A							0				
Dima and Frien	CF272-D2-A							0				
Nicholas and P	CF678-D2-A							0				
Toy Cars	CF545-D2-A							0				
DZY Loves Has	CF447-D2-A							0				
HQ9+	CF133-D2-A							0				
Holidays	CF670-D2-A							0				
Dividing Orange	CF244-D2-A							0				
Haiku	CF78-D2-A							0				
								0				
System of Equa	CF214-D2-A							0				
IQ test	CF25-D2-A							0				
Contest	CF501-D2-A							0				
Restoring Passw	CF94-D2-A							0				
Valera and Plat	CF369-D2-A							0				
Minimum Difficu	CF496-D2-A							0				
Little Elephant s	CF221-D2-A							0				
Collecting Beats	CF373-D2-A							0				
Letter	CF14-D2-A							0				
Kefa and First S	CF580-D2-A							0				
								0				
Ilya and Bank A	CF313-D2-A							0				
Uncowed Force	CF604-D2-A							0				
Reconnaissance	CF34-D2-A							0				
Lucky Ticket	CF146-D2-A							0				
Chat room	CF58-D2-A							0				
George and Sle	CF387-D2-A							0				
Ostap and Gras	CF735-D2-A							0				
The number of i	CF124-D2-A							0				
Table	CF359-D2-A							0				
Tavas and Nafa	CF535-D2-A							0				
								0				
Watermelon	CF4-D2-A							0				
Let's Watch Foc	CF195-D2-A							0				
Initial Bet	CF478-D2-A							0				
Saitama Destro	CF608-D2-A							0				
Queue on Bus s	CF435-D2-A							0				
Bicycle Chain	CF215-D2-A							0				
Little Elephant s	CF205-D2-A							0				
Amr and Music	CF507-D2-A							0				
Marks	CF152-D2-A							0				
Postcards and p	CF137-D2-A							0				
								0				
Business trip	CF149-D2-A							0				
Drazil and Date	CF515-D2-A							0				
Multiplication T	CF577-D2-A							0				
Exam	CF534-D2-A							0				
Alena's Schedu	CF586-D2-A							0				
Interview	CF631-D2-A							0				
Lucky Division	CF122-D2-A							0				
Appleman and I	CF462-D2-A							0				
Vasya and Digit	CF355-D2-A							0				
Parallelepiped	CF224-D2-A							0				
								0				
Group of Studen	CF357-D2-A							0				
Joysticks	CF651-D2-A							0				
Array	CF300-D2-A							0				
Round House	CF659-D2-A							0				
Lala Land and J	CF558-D2-A							0				
Autocomplete	CF53-D2-A							0				
Digital Counter	CF495-D2-A							0				
Vitaliy and Pie	CF525-D2-A							0				
Life Without Zer	CF75-D2-A							0				

	Problem Code	Status	Submit Count	Reading Time(m)	Thinking Time(m)	Coding Time(m)	Debug Time(m)	Total Time(m)	Problem Level /10	By yourself?	Category	1-2 line Comments About your approach
	AC Averages =>	0	0	0	0	0	0	0	0	0	0	0
								0				Watch - Thinking - Problem Simplification.
								0				Watch - Thinking - Brainstorm - Rank - Approach
								0				Study STL (You may study data structures if found hard)
								0				Watch - Combinatorics - Permutations and Combinations - 1
								0				Watch - Combinatorics - Permutations and Combinations - 2
	Petya and Country CF66-D2-B							0				Video Solution - Eng Muntaser Abukadeja
	Bear and Finding CF889-D2-B							9				Video Solution - Eng Muntaser Abukadeja
	Burglar and Match CF16-D2-B							\				Video Solution - Eng Muntaser Abukadeja
	Caisa and Pylo ns CF463-D2-B							0				Video Solution - Eng Muntaser Abukadeja
	Sum of Digits CF102-D2-B							0				Video Solution - Eng Muntaser Abukadeja
	Coins CF47-D2-B							0				Video Solution - Eng Samed Hajajla
	Effective Approach CF227-D2-B							0				Video Solution - Eng Abanob Ashraf
	Easter Eggs CF78-D2-B							0				Video Solution - Eng Abanob Ashraf
	Decoding CF746-D2-B							0				Video Solution - Solver to be (Java)
								0				Watch - Training-Secrets of Success
								0				Revise Stack/Queue datastructure concepts. Learn using STL
								0				Watch - Number Theory - Fib, GCD, LCM, Pow
	Big Mod UVA 374							0				
	Combinations UVA 369							0				
	Pi UVA 412							0				Video Solution - Eng Mohamed Adel
	Adding Reversed N UVA 713							0				Don't use big integer class. Write simple array computations
	Taxi TIMUS 1607							0				Can you get AC first submission?
	The Drunk Jailer LIVEARCHIVE 2557							0				Find a formula
	Vanya and Lantern CF492-D2-B							0				Video Solution - Solver to be (Java)
								0				Watch - Prefix Sum
	Kuriyama Mirai's S CF433-D2-B							0				
	Fence CF363-D2-B							0				Video Solution - Eng Muntaser Abukadeja
	President's Office CF6-D2-B							0				Video Solution - Eng Muntaser Abukadeja
	Lovely Palindrome CF688-D2-B							0				Video Solution - Solver to be (Java)
	Sort the Array CF451-D2-B							0				Video Solution - Solver to be (Java)
	Summer sell-off CF810-D2-B							0				Video Solution - Solver to be (Java)
	Colorful Field CF79-D12-B							0				Video Solution - Solver to be (Java)
	Keyboard CF88-D2-B							0				Video Solution - Eng Muntaser Abukadeja
	Mahmoud and a Tr CF766-D2-B							0				Video Solution - Solver to be (Java)
								0				
								0				Watch - Graph Theory - BFS
	Tic-Tac-Toe (I) SPOJ TOE1							0				Video Solution - Eng Ayman Salah
	Tic-Tac-Toe (II) SPOJ TOE2							0				Video Solution - Eng Essam AlNaggar
	Knight Moves UVA 439							0				Video Solution - Eng Magdy Hasan
	King's Path CF242-D2-C							0				Video Solution - Dr Mostafa Saad
	Bookworm TIMUS 1638							0				Can you get AC first submission
	UVA 10461							0				
	SPOJ POSTERIN							0				Sol
								0				
	Students and Shoe CF129-D2-B							0				Video Solution - Eng Abanob Ashraf
	Dreamoon and Wif CF476-D2-B							0				Video Solution - Eng Mohamed Adel
	Chat Online CF469-D2-B							0				Video Solution - Eng Mohamed Adel
	Olympic Medal CF215-D2-B							0				Video Solution - Eng Ahmed Salah
	Filya and Homework CF714-D2-B							0				Video Solution - Eng Muntaser Abukadeja
	Inna and New Matr CF400-D2-B							0				Video Solution - Eng Mohamed Salah
	Steps CF152-D2-B							0				Video Solution - Eng Muntaser Abukadeja
	Growing Mushroom CF186-D2-B							0				Video Solution - Eng Mohamed Salah
	Escape CF148-D2-B							0				Video Solution - Eng Ahmed Salah
								0				Review - Recursion
								0				Watch - Intro to DP - 1
								0				Watch - Intro to DP - 2
	Vacations CF699-D2-C							0				
	Woodcutters CF545-D2-C							0				
	Barcode CF225-D2-C							0				Video Solution - Dr Mostafa Saad
	Continents UVA 11094							0				Video Solution - Eng Ayman Salah
	Brownie Points UVA 10865							0				Video Solution - Eng Magdy Hasan
	Hanoi Tower TIMUS 1054							0				Sol
								0				
	Roma and Changir CF262-D2-B							0				Video Solution - Eng Mohamed Salah
	Bear and Strings CF385-D2-B							0				Video Solution - Eng Mohamed Salah
	I.O.U. CF376-D2-B							0				Video Solution - Eng Abanob Ashraf
	Jeff and Periods CF352-D2-B							0				Video Solution - Eng Muntaser Abukadeja
	Meeting CF144-D2-B							0				Video Solution - Eng Muntaser Abukadeja
	Chocolate CF617-D2-B							0				Video Solution - Eng Yahia Ashraf
	Easy Number Chal CF236-D2-B							0				Video Solution - Eng Yahia Ashraf
	Han Solo and Laze CF514-D2-B							0				
	Physics Practical CF253-D2-B							0				Video Solution - Eng Mohamed Salah
	Two Buttons CF520-D2-B							0				Video Solution - Solver to be (Java)
								0				
								0				Watch - Computational Geometry - Complex Number and 2D Point
								0				Watch - Computational Geometry - Lines and Distances
	Intersecting Lines UVA 378							0				
	The Stern-Brocot N UVA 10077							0				
	Mr. Kitayuta's Colo CF505-D2-B							0				Video Solution - Eng Muntaser Abukadeja
								0				
	DZY Loves Chemis CF445-D2-B							0				
	Kolya and Tanya CF584-D2-B							0				Video Solution - Eng Yahia Ashraf
	Suffix Structures CF448-D2-B							0				Video Solution - Eng Mohamed Salah
	Complete the Wor CF716-D2-B							0				Video Solution - Eng Mohamed Salah
	Sea and Islands CF544-D2-B							0				Video Solution - Eng Mohamed Salah
	Hopscotch CF141-D2-B							0				Video Solution - Eng Yahia Ashraf
	Valera and Contes CF369-D2-B							0				Video Solution - Eng Yahia Ashraf
	Bear and Friendshi CF791-D2-B							0				Video Solution - Eng Mohamed Salah
	Preparing Olympia CF550-D2-B							0				Video Solution - SolverToBe (Java)
								0				
								0				Watch - Focused and Diffused Thinking
								0				Watch - Graph Theory - MST - Kruskal
	Highways UVA 10147							0				Video Solution - Eng Mahmoud Adel
	ACM contest and E UVA 10600							0				Video Solution - Eng Moaz Rashad
	Virtual Friends UVA 11503							0				Video Solution - Eng Moaz Rashad
	Arctic Network UVA 10369							0				

	Problem Code	Status	Submit Count	Reading Time(m)	Thinking Time(m)	Coding Time(m)	Debug Time(m)	Total Time(m)	Problem Level /10	By yourself?	Category	1-2 line Comments About your approach
	AC Averages =>	0	0	0	0	0	0	0	0	0	0	0
Trees on the level	UVA 122							0				Video Solution - SolverToBe (Java)
Final Standings	TIMUS 1100							0				Stable sort exercise
Farm	TIMUS 1349							0				Learn Fermat's Last Theorem
								0				
Mashmikh and To	CF415-D2-B							0				Video Solution - Eng Salma Yehia
Approximating a C	CF602-D2-B							0				Video Solution - Eng Salma Yehia
Gena's Code	CF614-D2-B							0				
OR in Matrix	CF486-D2-B							0				
Fox And Two Dots	CF510-D2-B							0				Video Solution - Eng Mohamed Adel
Routine Problem	CF337-D2-B							0				Video Solution - Eng Mohamed Adel
Vasya and Wrestlin	CF493-D2-B							0				
Hamming Distance	CF608-D2-B							0				
Wet Shark and Bis	CF621-D2-B							0				Video Solution - Eng Mahmoud Mabrok
Kefa and Company	CF580-D2-B							0				Video Solution - SolverToBe (Java)
Tavas and SaDDa:	CF535-D2-B							0				Video Solution - Eng Abanob Ashraf
								0				
Minimum Ternary S	CF1009-D12-B							0				
	CF1030-D12-B							0				
	CF1051-D2-B							0				
	CF1237-D12-B							0				
								0				
								0				Watch - Intro to Greedy
Painting Eggs	CF282-D2-B							0				
Pasha Maximizes	CF435-D2-B							0				Video Solution - Eng Hossam Yehia
Little Girl and Gam	CF276-D2-B							0				Video Solution - Eng Hossam Yehia
Pasha and String	CF525-D2-B							0				Video Solution - Eng Hossam Yehia
Booking System	CF416-D2-C							0				
Vanya and Exams	CF492-D2-C							0				
The Skyline Proble	UVA 105							0				
Hanoi Tower Trout	UVA 10276							0				Video Solution - Eng Mahmoud Adel
Maze Exploration	UVA 784							0				Video Solution - Eng Mahmoud Adel
IP-TV	UVA 1174							0				
								0				Before moving to another sheet, email me with feedback about these problems selection.
								0				
								0				You don't have to or encouraged to solve the next problem. If you felt you need so, try some of them. Or Proceed to next and solve in parallel, up to you.
								0				
Devu, the Dumb G	CF439-D2-B							0				Video Solution - Solver to be (Java)
Find The Bone	CF796-D2-B							0				Video Solution - Solver to be (Java)
Regular Bracket St	CF26-D12-B							0				Video Solution - Solver to be (Java)
Inbox (100500)	CF465-D2-B							0				
Different is Good	CF672-D2-B							0				
Permutation	CF137-D2-B							0				
Little Elephant and	CF259-D2-B							0				
Airport	CF218-D2-B							0				
Commen --- The Be	CF732-D2-B							0				
Prison Transfer	CF427-D2-B							0				
A and B and Comp	CF519-D2-B							0				
Letter	CF43-D2-B							0				
Game of Robots	CF670-D2-B							0				
								0				
African Crossword	CF90-D2-B							0				
Cows and Poker G	CF284-D2-B							0				
Find Marble	CF285-D2-B							0				
Interesting drink	CF706-D2-B							0				
Megacity	CF424-D2-B							0				
Beautiful Paintings	CF651-D2-B							0				
Ilya and Queries	CF313-D2-B							0				
Code Parsing	CF255-D2-B							0				
Hungry Sequence	CF327-D2-B							0				
Chloe and the seq	CF743-D2-B							0				
Luxurious Houses	CF581-D2-B							0				
								0				
Settlers' Training	CF63-D2-B							0				
Far Relative's Prot	CF629-D2-B							0				
Wilbur and Array	CF596-D2-B							0				
Text Document An	CF723-D2-B							0				
Shower Line	CF431-D2-B							0				
Misha and Changin	CF501-D2-B							0				
Coat of Anticubism	CF667-D2-B							0				
Ternary Logic	CF136-D2-B							0				
Counting Rhombi	CF189-D2-B							0				
Pashmak and Flow	CF459-D2-B							0				
								0				
The Monster and th	CF592-D2-B							0				
The Fibonacci Seg	CF365-D2-B							0				
Spider Man	CF705-D2-B							0				
Little Robber Girl's	CF686-D2-B							0				
Unary	CF133-D2-B							0				
Canvas Frames	CF127-D2-B							0				
Ohana Cleans Up	CF554-D2-B							0				
Garland	CF408-D2-B							0				
Petya and Staircas	CF362-D2-B							0				
Equidistant String	CF545-D2-B							0				
Vanya and Food P	CF677-D2-B							0				
Calendar	CF304-D2-B							0				
Amr and Pins	CF507-D2-B							0				
Polo the Penguin a	CF289-D2-B							0				
								0				
George and Round	CF387-D2-B							0				
Alyona and flowers	CF740-D2-B							0				
Urbanization	CF735-D2-B							0				
Testing Pants for S	CF104-D2-B							0				
Cells Not Under At	CF701-D2-B							0				
Vanya and Books	CF552-D2-B							0				
Worms	CF474-D2-B							0				
Fortune Telling	CF59-D2-B							0				

	Problem Code	Status	Submit Count	Reading Time(m)	Thinking Time(m)	Coding Time(m)	Debug Time(m)	Total Time(m)	Problem Level /10	By yourself?	Category	1-2 line Comments About your approach
	AC Averages =>	0	0	0	0	0	0	0	0	0	0	0
Walking in the Rain	CF192-D2-B							0				
Dima and To-do List	CF366-D2-B							0				
Sail	CF298-D2-B							0				
Fox and Cross	CF389-D2-B							0				
Rebranding	CF591-D2-B							0				
Increase and Decrease	CF246-D2-B							0				
Alyona and Mex	CF682-D2-B							0				
Coins	CF58-D2-B							0				
Berland National List	CF567-D2-B							0				
Art Union	CF416-D2-B							0				
								0				
BerSU Ball	CF489-D2-B							0				
Random Teams	CF478-D2-B							0				
Friends	CF94-D2-B							0				
War of the Corporations	CF625-D2-B							0				
Road Construction	CF330-D2-B							0				
Binary Number	CF92-D2-B							0				
Before an Exam	CF4-D2-B							0				
Running Student	CF9-D2-B							0				
Anton and currencies	CF508-D2-B							0				
Phone Numbers	CF151-D2-B							0				

Problem Name	Problem Code	Status	Submit Count	Reading Time(m)	Thinking Time(m)	Coding Time(m)	Debug Time(m)	Total Time(m)	Problem Level /10	By yourself?	Category	1-2 line Comments About your approach
	AC Averages =>	0	0	0	0	0	0	0	0	0	0	0
												DON'T Skip colored problems. Don't skip others unless a block is really easy for you
												Remove the given link and write a comment. Start your comment with a classification for the problem: Useless, repeated idea, boring, normal, good problem, interesting problem or important problem.
Drazil and Factorial	CF515-D2-C							0				Video Solution - Dr Mostafa Saad
Lucky Permutation	CF304-D2-C							0				
Soldier and Cards	CF546-D2-C							0				
Watchmen	CF651-D2-C							0				
								0				
								0				Watch - Thinking - Concretely - Symbolically - Pictorially
								0				Watch - Thinking - Problem Constraints
								0				Watch - Number Theory - Primes
Fox Dividing Chees	CF371-D2-B							0				Video Solution - Eng Abanob Ashraf
Duff in Love	CF588-D2-B							0				
Twin Primes	UVA 10394							0				
Summation of Four	UVA 10168 [1]							0				Video Solution - Eng Moaz Rashad [2]
The Lottery	UVA 10325							0				Video Solution - Eng Amr Bahaa [3]
Hamburgers	CF371-D2-C							0				
Mint	UVA 10717							0				Sol
BITMAP - Bitmap	SPOJ BITMAP							0				
	CODECHEF GCDM							0				Sol uses __int128 to avoid overflow
	UVA 10843							0				Theory result to read
The Child and Set	CF 937-D2-B							0				
Tanya and Postcan	CF 318-D2-B							0				
								0				
Greg and Array	CF296-D2-C							0				
The World is a The	CF131-D2-C							0				Video Solution - Eng Youssef Ali
Trains	CF88-D2-C							0				Video Solution - Solver to be (Java)
Semifinals	CF 318-D2-B							0				
Towers	CF 478-D2-B							0				
								0				Watch - Algebra - Number Bases and Polynomials
To Carry or not to C	UVA 10469							0				Sol
Beat the Spread!	UVA 10812							0				
Summation of Poly	UVA 10302							0				
Polly the Polynomi	UVA 498							0				
	UVA 11053							0				Find O(n) Solution
	LiveArchive 8078							0				Sol
								0				Watch - Algebra - Patterns in Sequences
R U Kidding Mr. Fe	UVA 10509							0				
Wandering Queen	SPOJ QUEEN							0				Sol to read
Spreadsheet	UVA 196							0				
	HACKR sherlock-an							0				Sol
								0				Watch - Algebra - Summations
								0				Watch - Algebra - Basic Matrix Operations
Searching for Grap	CF402-D2-C							0				
Flying Saucer Segr	CF227-D2-C							0				
Vasya and Petya's	CF577-D2-C							0				
Round Table Knigh	CF71-D2-C							0				
Kefa and Park	CF580-D2-C							0				Video Solution - Solver to be (Java)
Knight Tournament	CF357-D2-C							0				
Special Offer! Supe	CF 318-D2-B							0				
								0				Watch - Thinking - Problem Abstraction
								0				Watch - Thinking - Problem Reverse
								0				Watch - Search Techniques - Backtracking
Graph Coloring	UVA 193							0				Video Solution - Dr Mostafa Saad
23 out of 5	UVA 10344							0				Video Solution - Eng Mohamed Nasser
8 Queens Chess P	UVA 750							0				Video Solution - Eng Ayman Salah
Assemble	UVA 12124							0				Sol
	SPOJ FUNPROB							0				Sol
								0				
Magic Formulas	CF424-D2-C							0				
Pythagorean Triple	CF707-D2-C							0				
Gerald's Hexagon	CF580-D2-C							0				
Points on Line	CF252-D2-C							0				
Find Maximum	CF353-D2-C							0				
Jzzhu and Sequenc	CF 106-D2-B							0				
Simple Game	CF 318-D2-B							0				
Prime Matrix	CF 371-D2-B							0				
								0				Review bitmasking
								0				Watch - DP - Subset Style
Vacation	UVA 10192							0				Explained in the tutorial videos
Dividing coins	UVA 562							0				Video Solution - Eng Ayman Salah
								0				Watch - DP - Consecutive Ranges Style
	SRM149-D1-500							0				
	SRM536-D2-1000							0				
The Blocks Problem	UVA 101							0				
Divisibility	UVA 10036							0				
	UVA 11628							0				Sol
								0				
Rational Resistanc	CF344-D2-C							0				
k-Multiple Free Set	CF275-D2-C							0				
Polycarpus' Dice	CF534-D2-C							0				Sol
Print Check	CF 937-D2-B							0				
Playing Cubes	CF 207-D2-B							0				
T-primes	CF 298-D2-B							0				
								0				Watch - DP - Nested Ranges Style
								0				Watch - DP - General Ranges Style
Creating Palindrom	UVA 11753							0				Video Solution - Eng Aya Elymany
Again Palindrome	UVA 10617							0				Sol to read
Exploring Pyramids	UVA 1362							0				Video Solution - Eng Ayman Salah
Cutting Sticks	UVA 10003							0				
Optimal Array Multi	UVA 348							0				Sol
Accordion Patience	UVA 127							0				Video Solution - Eng Moaz Rashad
Software CRC	UVA 128							0				Video Solution - Eng Moaz Rashad
	ZOJ 1200							0				
								0				

Problem Name	Problem Code	Status	Submit Count	Reading Time(m)	Thinking Time(m)	Coding Time(m)	Debug Time(m)	Total Time(m)	Problem Level /10	By yourself?	Category	1-2 line Comments About your approach
	AC Averages =>	0	0	0	0	0	0	0	0	0	0	0
Maze	CF378-D2-C							0				
Thor	CF705-D2-C							0				
Hard problem	CF706-D2-C							0				
Unusual Product	CF405-D2-C							0				
Palindrome Transf	CF486-D2-C							0				
Removing Column	CF496-D2-C							0				Video Solution - Dr Mostafa Saad
Crazy Town	CF499-D2-C							0				Video Solution - Dr Mostafa Saad
Queue	CF495-D2-B							0				
Vika and Squares	CF510-D2-B							0				
Cosmic Tables	CF528-D2-B							0				
								0				Watch - Thinking - Incrementally
								0				Watch - Thinking - Problem Domain re-interpretation
								0				Watch - Number Theory - Factorization
Prime Factors	UVA 583							0				
Mr. Azad and his S	UVA 10490							0				Sol to read
Perfect P-th Power	UVA 10622							0				Video Solution - Eng Moaz Rashad
Prime Land	UVA 516							0				
	UVA 10920							0				
	SRM274-D1-500							0				
								0				
Text Editor	CF253-D2-C							0				
Alternative Thinkin	CF604-D2-C							0				
Tennis Champions	CF735-D2-C							0				
Guess Your Way O	CF507-D2-C							0				Video Solution - Dr Mostafa Saad
Biathlon	CF84-D2-C							0				
Marina and Vasya	CF584-D2-C							0				
Divide by Three	CF792-D2-C							0				Video Solution - Solver to be (Java)
Covered Path	CF308-D2-B							0				
Facetook Priority W	CF76-D2-B							0				
Treasure Hunt	CF303-D2-B							0				
								0				
How Many Points	UVA 10790							0				Sol
Factovisors	UVA 10139							0				Sol to read
Fractions Again?!	UVA 10976							0				Sol to read
Cut Ribbon	CF189-D2-A							0				Video Solution - Solver to be (Java)
								0				
								0				Watch - Probability - First 9 videos
Cows and Cars	UVA 10491							0				Revise Probability
What is the Probab	UVA 10056							0				Sol
Let's Dance	UVA 10218							0				Sol
Probability(Given	UVA 11181							0				Sol
Another lottery	UVA 11628							0				Sol
Airplane	UVA 12461							0				Sol to read
	HACKR tower-3-col							0				Learn Fermat's little theorem
	CF445-D2-C							0				
	HACKR a-circle-and							0				
	UVA 11573							0				Learn 0/1 BFS
Rankings	UVA 12263							0				Sol
								0				
Hacker, pack your	CF822-D2-C							0				Video Solution - Solver to be (Java)
The Meaningless C	CF834-D2-C							0				Video Solution - Solver to be (Java)
Star sky	CF835-D2-C							0				Video Solution - Solver to be (Java)
								0				Before moving to another sheet, email me with feedback about these problems selection.
								0				
								0				You don't have to or encouraged to solve the next problem. If you felt you need so, try some of them. Or Proceed to next and solve in parallel, up to you.
								0				
Diverse Permutatio	CF483-D2-C							0				
Replacement	CF136-D2-C							0				
Homework	CF102-D2-C							0				
Little Elephant and	CF221-D2-C							0				
Developing Skills	CF581-D2-C							0				
Maxim and Discour	CF262-D2-C							0				
Fox and Box Accur	CF389-D2-C							0				
Ice Skating	CF218-D2-C							0				
Valera and Tubes	CF441-D2-C							0				
Secret	CF271-D2-C							0				
Key Task	SPOJ CERC07K							0				
Cleaning Robot	SPOJ CLEANRBT							0				
								0				
They Are Everywh	CF701-D2-C							0				
Monitor	CF16-D2-C							0				
System Administrat	CF22-D2-C							0				
Lucky Sum	CF122-D2-C							0				
NP-Hard Problem	CF688-D2-C							0				
Vladik and fractio	CF743-D2-C							0				
Case of Matryoshk	CF556-D2-C							0				
Vanya and Label	CF677-D2-C							0				
Exams	CF479-D2-C							0				
Boredom	CF456-D2-C							0				
Learning Language	CF278-D2-C							0				
Beautiful Sets of P	CF268-D2-C							0				
								0				
Strategic Defense	UVA 497							0				Explained in the tutorial videos
String to Palindrom	UVA 10739							0				Explained in the tutorial videos
Trouble of 13-Dots	UVA 10819							0				
Sagheer and Nubia	CF812-D2-C							0				Video Solution - Solver to be (Java)
Purification	CF330-D2-C							0				
Division into Team	CF149-D2-C							0				
Disposition	CF49-D2-C							0				
Mashmikh and Nu	CF415-D2-C							0				
Statues	CF129-D2-C							0				
Inna and Huge Car	CF400-D2-C							0				
Anagram Search	CF144-D2-C							0				
Ilya and Sticks	CF525-D2-C							0				
Day at the Beach	CF599-D2-C							0				

Problem Name	Problem Code	Status	Submit Count	Reading Time(m)	Thinking Time(m)	Coding Time(m)	Debug Time(m)	Total Time(m)	Problem Level /10	By yourself?	Category	1-2 line Comments About your approach
	AC Averages =>	0	0	0	0	0	0	0	0	0	0	0
								0				
Appleman and Toa	CF462-D2-C							0				Sol
Anya and Smartph	CF518-D2-C							0				
Little Girl and Maxir	CF276-D2-C							0				
Sereja and Algori	CF368-D2-C							0				
The Child and Toy	CF437-D2-C							0				
Perfect Pair	CF318-D2-C							0				
Another Problem o	CF165-D2-C							0				
Socks	CF731-D2-C							0				
Valera and Election	CF369-D2-C							0				

Problem Name	Problem Code	Status	Submit Count	Reading Time(m)	Thinking Time(m)	Coding Time(m)	Debug Time(m)	Total Time(m)	Problem Level /10	By yourself?	Category	1-2 line Comments About your approach
	AC Averages =>	0	0	0	0	0	0	0	0	0	0	0
								0				Watch - Thinking - Search Space and Output Analysis
								0				Watch - Thinking - Observations Discovery
								0				Watch - Game Theory - Intro
Win or Freeze	CF151-D2-C							0				Video Solution - Dr Mostafa Saad
Euclid's Game	UVA 10368							0				Video Solution - Eng Moaz Rashad
Pyramids	SPOJ PIR							0				Sol
Power of Cryptograph	UVA 113							0				Sol to read
	SRM458-D2-500							0				
Is There A Second Value	UVA 10462							0				
	SRM381-D2-1000							0				
								0				
Modified GCD	CF75-D2-C							0				Video Solution - Dr Mostafa Saad
Alyona and mex	CF740-D2-C							0				Video Solution - Dr Mostafa Saad
Hamburgers	CF371-D2-C							0				
Wet Shark and Flow	CF621-D2-C							0				
Predict Outcome of Fight	CF451-D2-C							0				
Balls and Boxes	CF260-D2-C							0				Video Solution - Dr Mostafa Saad
Alice and Bob	CF347-D2-C							0				Video Solution - Eng Mohamed Nasser
Mahmoud and Ehab	CF959-D2-C							0				Video Solution - Eng Mohamed Salah
Almost Equal	CF1206-D2-C							0				Video Solution - Dr Mostafa Saad
	CF1220-D12-C							0				
	CF1065-D2-C							0				
	CF1036-D2-C							0				
	CF1068-D2-C							0				
	CF313-D2-C							0				
Balls Game	CF430-D2-B							0				
Magical Array	CF266-D2-B							0				
								0				Watch - Thinking - Misc - Solution Verification - Implementation
								0				Watch - Graph Theory - Dijkstra
Jugs	UVA 571							0				Video Solution - Dr Mostafa Saad
Sending email	UVA 10986							0				
Lift Hopping	UVA 10801							0				
Shopping	SPOJ SHOP							0				
Ordering	UVA 872							0				
	CF1084-D2-C							0				
	CF1059-D2-C							0				
	CF101933-GYM-K							0				Sol
Karen and Coffee	CF816-D2-B							0				Video Solution - Dr Mostafa Saad
								0				
Bulls and Cows	CF63-D2-C							0				Sol
Xor-tree	CF430-D2-C							0				
Median Smoothing	CF591-D2-C							0				
Coloring Trees	CF711-D2-C							0				Video Solution - Solver to be
Clear Symmetry	CF202-D2-C							0				
	CF1237-D12-C2							0				
Sereja and Mirroring	CF426-D2-B							0				
Restoring Painting	CF275-D2-B							0				
								0				
								0				Watch - Computational Geometry - Lines Intersections
Gleaming the Cubes	UVA 737							0				Sol
Intersecting Line Segments	UVA 866							0				Sol
								0				Watch - Computational Geometry - Circles
The Circumference of Circle	UVA 438							0				Sol
Points in Figures: Rectangles	UVA 477							0				Sol
Square Pegs And Rectangles	UVA 356							0				Sol to read
	UVA 453							0				Learn Handling Precisions
Divisibility of Factors	UVA 10484							0				Sol to read
	SRM436-D2-500							0				
	CF975-D2-C							0				
	CF1047-D2-C							0				
	CF1075-D2-C							0				
	CF758-D2-C							0				
	UVA 10525							0				Video Sol. Also solvable in 2 other ways.
								0				
Prime Permutation	CF124-D2-C							0				
Hometask	CF155-D2-C							0				
Terse princess	CF148-D2-C							0				Video Solution - Eng Mohamed Nasser
Hacking Cypher	CF490-D2-C							0				
Dreamoon and Sum of Digits	CF476-D2-C							0				Video Solution - Dr Mostafa Saad
Try and Catch	CF195-D2-C							0				Editorial - Eng Ahmed Osama
Primes or Palindromes	CF569-D2-C							0				
View Angle	CF257-D2-C							0				Editorial - Eng Ahmed Osama
Little Pony and Sort by Shift	CF559-D2-B							0				
								0				Watch - Thinking - Error Inspection - History - Contest Strategy
								0				Watch - DP - Building Output
Unidirectional TSP	UVA 116							0				
Make Palindrome	UVA 10453							0				Sol
Fast Food	UVA 662							0				
Palindromic Subsequences	UVA 11404							0				
Gone Fishing	UVA 757							0				Sol to read
Special Olympics	CF199-D2-B							0				
Rings and Glue	UVA 10301							0				Sol
								0				Watch - DP - Counting
k-Tree	CF431-D2-C							0				Video Solution - Solver to be (Java)
Caesar's Legions	CF118-D2-D							0				
UnsealTheSafe	SRM354-D2-1000							0				
DiceGames	SRM349-D1-500							0				
	SPOJ TWINSNOW							0				Sol - text clarification
	SPOJ FACENEMY							0				Sol
								0				
No to Palindromes!	CF465-D2-C							0				
Triangle	CF408-D2-C							0				
To Add or Not to Add	CF231-D2-C							0				
Number of Ways	CF466-D2-C							0				Video Solution - Solver to be (Java)
Queue	CF141-D2-C							0				
Magical Boxes	CF270-D2-C							0				

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	AC Averages =>	0	0	0	0	0	0	0	0	0	0	0
Find Pair	CF160-D2-C							0				
Multitasking	CF388-D2-B							0				
Non-square Equation	CF333-D2-B							0				
								0				Watch - Thinking - Let's Put All Together
								0				Watch - DP - Table Method
								0				Watch - Graph Theory - Floyd Warshal
Frogger	UVA 534							0				Sol
Identifying Concurrency	UVA 334							0				
Numbering Paths	UVA 125							0				Sol
Jack Straws	UVA 273							0				Sol
Longest Match	UVA 10100							0				
Isolated Segments	UVA 11343							0				Sol
Counting	UVA 10198							0				Needs Big Integer: Have it in your c++ library or learn Java for these (rare) cases
								0				
Mafia	CF349-D2-C							0				
Sereja and Prefixes	CF381-D2-C							0				
About Bacteria	CF199-D2-C							0				
DNA Alignment	CF520-D2-C							0				
Geometric Progression	CF567-D2-C							0				
Watering Flowers	CF617-D2-C							0				
Quiz	CF337-D2-C							0				
Secret Combination	CF398-D2-B							0				
MUH and Important	CF401-D2-B							0				
Lucky Mask	CF346-D2-B							0				
								0				
								0				Watch - Measuring Algorithms Performance - 2
								0				Watch - Graph Theory - Tree Diameter and Isomorphism
PT07Z	SPOJ PT07Z							0				Sol
Roads in the North	UVA 10308							0				Sol
Subway tree system	LIVEARCHIVE 2935							0				Sol
								0				
Shaass and Lights	CF294-D2-C							0				Video Solution - Dr Mostafa Saad
Journey	CF721-D2-C							0				
Captain Marmot	CF474-D2-C							0				Video Solution - Dr Mostafa Saad
The Big Race	CF592-D2-C							0				
Molly's Chemicals	CF778-D2-C							0				Video Solution - Solver to be (Java)
Anatoly and Cockroaches	CF118-D2-B							0				
Opposites Attract	CF331-D2-B							0				
								0				
Railway	UVA 10263							0				Sol to read
Factorial Factors	UVA 884							0				
Wifi Access	UVA 12748							0				Sol
Lining Up	UVA 270							0				Video Solution - Eng Mohamed Nasser. Don't Code O(N^3)
Pouring water	SPOJ POUR1							0				Video Solution - Eng Moaz Rashad
	CF23-D12-C							0				
	CF869-D2-C							0				
	SRM321-D1-500							0				See Rushiose's code in arena summary
								0				
								0				Watch Video - Expected Value
God, Save me	UVA 10777							0				Sol
	CF839-D2-C							0				
	CF454-D2-C							0				
	SRM577-D1-250							0				Editorial
	HACKR lazy-sorting							0				Revise Expected Value
	SPOJ ALIENS							0				Sol - Practice on min enclosing circle
	CF340-D2-B							0				
								0				Before moving to another sheet, email me with feedback about these problems selection.
								0				
								0				Optional Problems
								0				You don't have to or encouraged to solve the next problem. If you felt you need so, try some of them. Or Proceed to next and solve in parallel, up to you.
								0				
Checkposts	CF427-D2-C							0				
Literature Lesson	CF139-D2-C							0				
Arpa's loud Owf and	CF742-D2-C							0				
Parity Game	CF298-D2-C							0				
Beauty Pageant	CF246-D2-C							0				
Heroes	CF80-D2-C							0				
Dynasty Puzzles	CF192-D2-C							0				
Buns	CF106-D2-C							0				
Counting Kangaroos	CF373-D2-C							0				
Corporation Mail	CF56-D2-C							0				
Matrix	CF365-D2-C							0				
Pick up sticks	UVA 11686							0				Sol
								0				
Little Elephant and Integers	CF205-D2-C							0				
Sereja and Contest	CF315-D2-C							0				
Vasya and Robot	CF355-D2-C							0				
Hockey	CF86-D2-C							0				
Petya and File System	CF66-D2-C							0				
Kyoya and Colored Flowers	CF554-D2-C							0				
George and Job	CF467-D2-C							0				
Harmony Analysis	CF610-D2-C							0				
Anton and Making Pairs	CF734-D2-C							0				
Table Decorations	CF478-D2-C							0				
Recycling Bottles	CF672-D2-C							0				
								0				
Message	CF157-D2-C							0				
Wilbur and Points	CF596-D2-C							0				
Cows and Sequence	CF284-D2-C							0				
Ladder	CF279-D2-C							0				
Not Wool Sequence	CF239-D2-C							0				
Anagram	CF254-D2-C							0				
DZY Loves Sequences	CF447-D2-C							0				
DZY Loves Physics	CF445-D2-C							0				
Misha and Forest	CF501-D2-C							0				
Jzzhu and Chocolate	CF450-D2-C							0				

Problem Name	Problem Code	Status	Submit Count	Reading Time(m)	Thinking Time(m)	Coding Time(m)	Debug Time(m)	Total Time(m)	Problem Level /10	By yourself?	Category	1-2 line Comments About your approach
	AC Averages =>	0	0	0	0	0	0	0	0	0	0	0
Cinema	CF670-D2-C							0				
								0				
Report	CF631-D2-C							0				
Bear and Prime Num	CF385-D2-C							0				
Robbery	CF90-D2-C							0				
Vasya and Basketba	CF493-D2-C							0				
Vanya and Scales	CF552-D2-C							0				
Pashmak and Buses	CF459-D2-C							0				
Fancy Number	CF118-D2-C							0				

Problem Name	Problem Code	Status	Submit Count	Reading Time(m)	Thinking Time(m)	Coding Time(m)	Debug Time(m)	Total Time(m)	Problem Level /10	By yourself?	Category	1-2 line Comments About your approach
	AC Averages ==>	0	0	0	0	0	0	0	0	0	0	0
Dividing Island	CF63-D2-D							0				
Flowers	CF474-D2-D							0				Video Solution - Solver to be (Java)
Dima and Bacteria	CF400-D2-D							0				
	CF1043-D12-C							0				
	CF1033-D12-C							0				
	CF1066-D3-E							0				
	CF534-D2-D							0				
	CF899-D2-E							0				
	CF729-D12-D							0				
Tourist Problem	CF398-D2-C							0				
Lorenzo Von Matter	CF381-D4-E							0				
Restore Graph	CF349-D3-C							0				
	CF309-D1-C							0				
	CF101-D1-B							0				Sol
	SRM569-D2-1000							0				
	CF961-D12-D							0				
	CF955-D2-C							0				
	UVA 12869							0				Sol
	CF372-D1-B							0				
								0				Watch - Data Structures - Segment Tree (2 vid)
Interval Product	UVA 12532							0				
Potentiometers	LIVEARCHIVE 2191							0				
Halt The War	SPOJ CDC12_H							0				
Counting Primes	SPOJ CNTPRIME							0				
Horrible Queries	SPOJ HORRIBLE							0				
Light Switching	SPOJ LITE							0				
Circular RMQ	CF52-D12-C							0				
A Famous City	SPOJ CITY2							0				Sol
RMQ with Shifts	UVA 12299							0				See sscanf and sprintf usage
R2D2 and Droid Army	CF514-D2-D							0				Use rmq
Ahoy, Pirates!	UVA 11402							0				Sol
Brackets	SPOJ BRCKTS							0				Sol
Present	CF460-D2-C							0				
MessageMess	SRM149-D1-500							0				
DiceGames	SRM349-D1-500							0				
Mirror, Mirror	UVA 466							0				
Maximum Sum	SPOJ KGSS							0				
	SRM297-D1-500							0				
	SRM441-D1-250							0				
	CF201-D1-B							0				
	CF380-D1-C							0				
	CF161-D12-D							0				Reading: DP on Trees
	CF61-D2-E							0				
	SPOJ KOMPICI							0				
								0				
Quantity of Strings	CF151-D2-D							0				
Eternal Victory	CF61-D2-D							0				
Array Division	CF808-D2-D							0				Video Solution - Solver to be (Java)
	CF45-D12-D							0				
	SRM428-D2-1000							0				
	SGU 321							0				Sol
	CODECHEF OPPOSIT							0				
	SRM513-D2-1000							0				
	SRM292-D1-500							0				
	SRM405-D2-1000							0				
Hiring Staff	CF218-D2-C							0				
Tavas and Karafs	CF339-D4-E							0				
Permutations	CF186-D3-C							0				Sol
								0				Watch - Two pointers technique
Spider's Web	CF216-D2-D							0				
Chips	CF334-D2-D							0				
Vasya and String	CF676-D2-C							0				
The SetStack Comp	LiveArchive 3634							0				Sol
Database	UVA 1592							0				
Can you answer the	SPOJ GSS1							0				Sol
	SPOJ BILLIARD							0				Sol
Can you answer the	SPOJ GSS3							0				
	SPOJ ABA12E							0				Sol
	UVA 11825							0				Sol
	CF472-D12-D							0				
	UVA 12325							0				Prove your Solution
	UVA 12047							0				Sol
	UVA 10705							0				Sol
	UVA 1555							0				Sol
	CF80-D2-D							0				
								0				
Mahmoud and a Dict	CF766-D2-D							0				Video Solution - Solver to be (Java)
An overnight dance	CF814-D2-D							0				Video Solution - Solver to be (Java)
Polyline	CF617-D2-D							0				
Queue	CF92-D2-D							0				
	CF1038-D2-D							0				
	CF552-D2-D							0				
	CF101917-D12-E							0				
	CF1058-D2-D							0				
	CF1042-D12-D							0				
	SPOJ BIA							0				Sol
Plant	CF198-D2-C							0				
Reberland Linguistic	CF381-D4-E							0				
Lucky Permutation	CF287-D3-C							0				DP - Probability
								0				
Dice Throwing	UVA 10759							0				Sol
TestBettingStrategy	SRM339-D1-500							0				
Collecting Bugs	PKU 2096							0				Sol
France '98	UVA 542							0				Sol

Problem Name	Problem Code	Status	Submit Count	Reading Time(m)	Thinking Time(m)	Coding Time(m)	Debug Time(m)	Total Time(m)	Problem Level /10	By yourself?	Category	1-2 line Comments About your approach
	AC Averages ==>	0	0	0	0	0	0	0	0	0	0	0
Tribbles	UVA 11021							0				Sol
Tennis contest	UVA 12457							0				Sol
Water Falls	UVA 833							0				
Number Sequence	UVA 10706							0				
Is It A Tree?	UVA 615							0				
Help R2-D2!	SPOJ.HELPR2D2							0				
	CF1016-D2-E							0				
	UVA 11997							0				Sol
	FbHkrCup 18-R1-A							0				
	SRM456-D2-1000							0				
								0				
Andrey and Problem	CF443-D2-D							0				Sol
Three Logos	CF581-D2-D							0				
Good Sequences	CF265-D2-D							0				
Party	CF135-D2-E							0				
Cupboard and Balloons	CF346-D2-E							0				
Cycles	CF236-D2-E							0				
								0				DP - Masks (2 vid)
Pebble Solitaire	UVA 10651							0				
Kefa and Dishes	CF580-D2-D							0				Video Solution - Solver to be
Permutations	SPOJ.PERMUT1							0				
Assignments	SPOJ.ASSIGN							0				
	CF16-D2-E							0				
Count the Faces.	UVA 10178							0				Read first Euler Formula
LCM Cardinality	UVA 10892							0				
Robot Rapping Res	CF645-D12-D							0				
Wavio Sequence	UVA 10534							0				Sol
	UVA 10342							0				Sol - read the statement clarification
								0				
Directed Roads	CF711-D2-D							0				
Block Tower	CF327-D2-D							0				
A and B and Interes	CF519-D2-D							0				
As Fast As Possible	CF701-D2-D							0				
Chloe and pleasant	CF743-D2-D							0				
Roads in Berland	CF265-D2-E							0				
Photographer	CF266-D2-E							0				
LCM Challenge	CF236-D2-E							0				
								0				String Processing - Trie
Search in the diction	SPOJ.DICT							0				
Disk Tree	UVA 1558							0				
Phone List	SPOJ.PHONELST							0				
Cellphone Typing	UVA 12526							0				
Vasily's Multiset	CF706-D2-D							0				
Exchange Rates	UVA 10113							0				
Equation	UVA 727							0				
Safe	CF47-D2-D							0				
Central Post Office	UVA 12379							0				Sol
Permalex	UVA 153							0				Sol
								0				DP - Sub-rectangle style
	UVA 507							0				
	UVA 10667							0				
								0				
Volleyball	CF96-D2-D							0				
Lazy Student	CF606-D2-D							0				
Multiplication Table	CF448-D2-D							0				Video Solution - Solve to be (Java)
	CF486-D2-D							0				
	CF1040-D2-D							0				
	CF264-D1-C							0				
	CF506-D1-A							0				
	CODECHEF.KSUM							0				
	CF623-D1-B							0				
Divisible by Seven	CF235-D2-E							0				
Devu and Partitionin	CF506-D2-E							0				
Arthur and Table	CF293-D2-E							0				

Problem Name	Problem Code	Status	Submit Count	Reading Time(m)	Thinking Time(m)	Coding Time(m)	Debug Time(m)	Total Time(m)	Problem Level /10	By yourself?	Category	1-2 line Comments About your approach
	AC Averages =>	0	0	0	0	0	0	0	0	0	0	0
								0				String Processing - KMP (2 vid)
Oulipo	PKU 3481							0				
A Needle in the Ha	SPOJ NHAY							0				
Finding the Tesser	SPOJ TESSER							0				
Period	SPOJ PERIOD							0				
Prefixes and Suffix	CF432-D2-D							0				
Tavas and Maleka	CF535-D2-D							0				
Be Efficient	UVA 11155							0				
Vertex Cover	SPOJ PT07X							0				Sol
First Digit Law	CF54-D12-C							0				
	CF500-D12-D							0				
	HACKR vertical-sticks							0				
	UVA 10174							0				
	UVA 1333							0				Sol - Text/Background Clarification
	CF842-D2-D							0				
	CF709-D2-D							0				
	SPOJ MSKYCODE							0				Sol
	LiveArchive 8015							0				Sol
								0				
Robin Hood	CF672-D2-D							0				
End of Exams	CF94-D2-D							0				
Equivalent Strings	CF560-D2-D							0				Sol to learn
Count Good Subst	CF451-D2-D							0				
Mushroom Scientis	CF188-D2-D							0				
Analyzing Polyline	CF195-D2-D							0				
	CF1023-D12-E							0				
	CF1060-D12-C							0				
Bear and Prime 10	CF980-D2-C							0				
Team	CF401-D2-C							0				
								0				DP - Games (2 vid)
Bachet's Game	UVA 10404							0				Sol
EllysCheckers	SRM534-D1-250							0				
RowAndCoins	SRM522-D1-250							0				
BagsOfGold	SRM228-D1-500							0				
Bag of mice	CF148-D2-D							0				
	CF1147-D1-B							0				
MELE3	SPOJ MELE3							0				Sol
Roads	SPOJ ROADS							0				Sol
The Tree Root	UVA 10459							0				Sol
SKYLINE	UVA 1232							0				Sol
Ordering the Soldi	SPOJ ORDERS							0				Sol
Playlist	CF268-D2-E							0				Sol
	SRM481-D1-500							0				
								0				
Little Girl and Maxi	CF278-D2-D							0				See editorials
Two Strings	CF224-D2-D							0				Sol
Big Maximum Sum	CF75-D2-D							0				
	SPOJ BROKTS2							0				Sol
	CF1057-D12-C							0				
	CF1066-D3-F							0				
	CF1064-D2-E							0				
	CF459-D2-E							0				
	UVA 10888							0				
	CF1043-D12-D							0				
Football Champion	CF205-D2-C							0				
Given Length and	CF495-D2-C							0				
								0				
Trip Routing	UVA 186							0				Sol
Scheduling Lectur	UVA 607							0				Sol
Weird Function	SPOJ WEIRDFN							0				Sol
The ? 1 ? 2 ? ... ?	UVA 10025							0				
Dictionary Subseq	SPOJ DICTSUB							0				Sol
Jimmi's Riddles	UVA 10058							0				Sol
Friends and Subse	CF689-D2-D							0				
Sum of Squares w	SPOJ SEGSQLRSS							0				Sol
Travel in Desert	UVA 10816							0				Sol
Almost Union-Find	UVA 11987							0				Sol
	SRM537-D2-1000							0				
	CF513-D12-C							0				Sol
	SRM453.5-D2-1000							0				
	SPOJ PARSUMS							0				Sol
	CF1138-D2-D							0				
								0				
Cow Program	CF284-D2-D							0				
Random Task	CF431-D2-D							0				
Greg and Graph	CF296-D2-D							0				
Russian Roulette	CF104-D2-D							0				
Bicycle Race	CF659-D2-D							0				
Greenhouse Effect	CF270-D2-D							0				
	CF645-D12-D							0				
	CF459-D2-C							0				
	CODECHEF REDCGA							0				
	CF1005-D3-F							0				
Pocket Book	CF158-D2-C							0				
Levko and Array R	CF361-D2-C							0				
Ice Cave	CF296-D2-C							0				
								0				
Robbery	UVA 707							0				Sol
The Errant Physi	UVA 126							0				Sol
Brackets sequence	UVA 1626							0				Sol
Unique World	UVA 10448							0				Video Solution - Dr Mostafa Saad
Bad Luck Island	CF540-D2-D							0				
Shopping Trip	UVA 11284							0				Sol
Hotel booking	UVA 11635							0				Sol
	CF337-D2-D							0				Sol
	HACKR ajourney							0				

Problem Name	Problem Code	Status	Submit Count	Reading Time(m)	Thinking Time(m)	Coding Time(m)	Debug Time(m)	Total Time(m)	Problem Level /10	By yourself?	Category	1-2 line Comments About your approach
	AC Averages =>	0	0	0	0	0	0	0	0	0	0	0
	CF665-D12-E							0				
								0				
Hit Ball	CF203-D2-D							0				
Sereja ans Anagra	CF368-D2-D							0				Sol
Choosing Capital f	CF219-D2-D							0				
Coloring Brackets	CF149-D2-D							0				Sol
Cycle in Graph	CF263-D2-D							0				
	CF101187-GYM-F							0				Sol
	SRM319-D1-500							0				
	AtCoder092-ARC-B							0				
	AtCoder002-AGC-C							0				
Fixing Typos	CF363-D1-C							0				
Cutting Figure	CF196-D1-C							0				
Escape from Stone	CF265-D1-C							0				
								0				
								0				Geometry - Simple and Convex Polygons
								0				Geometry - Polygon Area - Centroid - Cut
BestTriangulation	SRM278-D2-500							0				
Trees on My Island	UVA 10088							0				
Packing polygons	UVA 10005							0				Sol
	LIVEARCHIVE 2831							0				Use polygon cut
Video Surveillance	UVA 588							0				Use polygon cut
	SRM514-D1-500							0				
	SRM473-D1-500							0				
	SRM555-D2-1000							0				
	UVA 557							0				Sol
	SRM285-D1-500							0				
Xrange's Pancake	HACKR xrange-and-piz							0				Sol
	SRM525-D1-500							0				
	UVA 11648							0				Sol
	CF101864-GYM-A							0				Sol
	CF101864-GYM-L							0				Sol
	CF28-D12-C							0				
								0				Geometry - Point in polygon
	UVA 881							0				Sol
	UVA 11665							0				Sol
	TIMUS 1599							0				Sol
Polygons	UVA 137							0				Sol
								0				Graph Theory - Maximum Flow (2 vid)
Potholes	SPOJ POTHOLE							0				Sol
Power Transmissi	UVA 10330							0				Sol
Gopher II	UVA 10080							0				Sol
Software Allocator	UVA 259							0				Sol
	UVA 10349							0				Sol - 2 ways
	UVA 12168							0				Sol
A Plug for UNIX	UVA 753							0				Sol
	UVA 10349							0				Sol - 2 ways
Intergalactic Map	SPOJ IM							0				Sol
	UVA 11159							0				Sol
	UVA 1194							0				Sol
Fence Obstacle C	PKU 2374							0				Sol
River Crossing	UVA 10514							0				Sol
	SRM368-D1-500							0				Sol
	SRM373-D2-1000							0				Sol
	SRM558-D1-250							0				
	ZOJ 2587							0				Sol
	SRM550-D2-1000							0				
	UVA 10180							0				Sol
	TIMUS 1156							0				
	UVA 1184							0				Sol
	UVA 670							0				Sol

Problem Name	Problem Code	Status	Submit Count	Reading Time(m)	Thinking Time(m)	Coding Time(m)	Debug Time(m)	Total Time(m)	Problem Level /10	By yourself?	Category	1-2 line Comments About your approach
	AC Averages =>	0	0	0	0	0	0	0	0	0	0	0
								0				Graph Theory - SCC (2 vid)
The Bottom of a G	SPOJ BOTTOM							0				Sol
Test	UVA 10731							0				Sol
	SRM312-D1-500							0				
	CF467-D2-D							0				
Theseus and labyr	CF676-D2-D							0				
Cunning Gena	CF418-D1-B							0				
Sabotage	UVA 10480							0				Sol
	SRM352-D2-1000							0				
Garland	UVA 1555							0				Sol
	CF101589-GYM-F							0				Sol
	CF1016-D12-D							0				
	CF26-D12-D							0				Sol - must read
	CF1012-D1-B							0				
	CF1010-D1-C							0				
	CF833-D12-D							0				
	HACKR house-location							0				Sol
	CF621-D2-D							0				Sol
	CF101992-GYM-D							0				Sol
	SRM608-D2-1000							0				Sol
								0				
Gifts by the List	CF681-D2-D							0				
DZY Loves Modifi	CF447-D2-D							0				Prove
Mike and Feet	CF548-D2-D							0				
Special Grid	CF435-D2-D							0				
Roman and Numb	CF401-D2-D							0				
Persistent Bookca	CF707-D2-D							0				Sol
Regular Bridge	CF550-D2-D							0				
	CF1059-D2-D							0				
Almost Arithmetica	CF 325-D2-C							0				
Title	CF 386-D2-C							0				
Treasure	CF 390-D2-C							0				
								0				
Unique Attack	ZOJ 2587							0				
End of Fun	SPOJ DCEPC12E							0				
Grammar Evaluati	UVA 622							0				Sol
Find the Winning A	UVA 10111							0				Sol
Check the difficulty	PKU 2151							0				Sol
Proving Equivalen	UVA 12167							0				Sol
DDF	UVA 547							0				
Dominos	UVA 11504							0				Sol
	SRM419-D2-1000							0				
Winning Streak	UVA 11176							0				Sol
	SRM391-D2-1000							0				
	SRM465-D1-500							0				Sol
	UVA 10740							0				Sol
	UVA 12261							0				
	LIVEARCHIVE 4008							0				
	UVA 1342							0				Sol
	CF811-D2-D							0				
	AtCoder026-AGC-B							0				Sol
	SPOJ FISHES							0				Sol
	UVA 11475							0				Sol
								0				
Red-Green Tower	CF478-D2-D							0				
Renting Bikes	CF363-D2-D							0				
Lucky Number 2	CF146-D2-D							0				
Tennis Game	CF496-D2-D							0				
Bubble Sort Graph	CF340-D2-D							0				
Upgrading Array	CF402-D2-D							0				
	ZOJ 3305							0				Sol
	CF1017-D12-D							0				
Game	CF 386-D2-C							0				
Ciel and Robot	CF 325-D2-C							0				
Plus and Square R	CF 325-D2-C							0				
								0				
Boxes in a Line	UVA 12657							0				Sol
	SPOJ QUEST4							0				Sol
Multifactorials	UVA 11347							0				
Crimewave	UVA 563							0				Sol
	SRM545-D2-1000							0				
	SRM495-D1-500							0				
Primitive Root	SPOJ PROOT							0				Sol
Of Zorcs and Axes	CF101149-GYM-G							0				Sol
Connected Comp	CF292-D12-D							0				
AND Rounds	SPOJ ANDROUND							0				Sol
Campus Roads	UVA 11473							0				Sol
The Child and Zoo	CF437-D2-D							0				Sol
	CF403-D1-C							0				
	CF787-D2-C							0				
	CF309-D12-B							0				
	SRM392-D1-1000							0				
	UVA 12128							0				
	Timus 1362							0				Sol
	CF1012-D1-C							0				
	SPOJ COCONUTS							0				Sol
	FbHkrCup 18-RQ-C							0				
	LIVEARCHIVE 4682							0				Sol
								0				
Image Preview	CF651-D2-D							0				
Maximum Xor Sec	CF281-D2-D							0				
Ilya and Roads	CF313-D2-D							0				
Mr. Bender and Sc	CF255-D2-D							0				
Fish Weight	CF298-D2-D							0				
T-decomposition	CF237-D2-D							0				
Wizards and Huge	CF168-D2-D							0				

Problem Name	Problem Code	Status	Submit Count	Reading Time(m)	Thinking Time(m)	Coding Time(m)	Debug Time(m)	Total Time(m)	Problem Level /10	By yourself?	Category	1-2 line Comments About your approach
	AC Averages =>	0	0	0	0	0	0	0	0	0	0	0
	CODECHEF BJUDGE							0				
Dima and Salad	CF 369-D2-C							0				
								0				
Arbitrage	UVA 104							0				Sol
Random Task	CF431-D2-D							0				
Black Box	UVA 501							0				Sol - Must Read
Expressions	UVA 11234							0				Sol
Showstopper	SPOJ MSE07E							0				Read SPOJ users' comments about IO. See here sol
Tobo or not Tobo	SPOJ ANARC08A							0				Sol
Sum-up the Prime:	UVA 10419							0				Sol
Largest Rectangle	SPOJ HISTOGRAM							0				Sol. Don't implement as adhoc/greedy/Pure STL. Use a data structure.
	UVA 663							0				Sol
KingdomReorganize	SRM531-D2-1000							0				
The Problem with I	UVA 10092							0				
Psycho	SPOJ.PSYCHON							0				
Minimal Ratio Tree	LIVEARCHIVE 4326							0				
RACING	UVA 1234							0				Sol
ActivateGame	SRM470-D2-1000							0				
Pair of Numbers	CF359-D2-D							0				Sol
Nuts for nuts	UVA 10944							0				
Probability	UVA 11346							0				Sol
	SRM470-D1-500							0				
	SPOJ COCONUTS							0				Sol
	CF592-D2-D							0				
	UVA 1218							0				Sol
	SPOJ IOPC1207							0				Sol
	CF867-D12-E							0				
								0				
AlgoRace	CF189-D2-D							0				Sol
Modular Arithmet	CF604-D2-D							0				Sol
Lucky Transform	CF122-D2-D							0				
Boring Partition	CF239-D2-D							0				Sol. Find proof (See editorial comments)
Spongebob and St	CF599-D2-D							0				
How many trees?	CF9-D2-D							0				
	CF1043-D12-E							0				
	UVA 10982							0				Sol
	CF1060-D12-D							0				
Chulhu	CF 104-D2-C							0				
Anya and Ghosts	CF 308-D2-C							0				
Square Subsets	CF 368-D2-C							0				
								0				
Angry Programme	UVA 11506							0				Sol
The New Rule in E	UVA 10742							0				Sol
Multiples of 3	SPOJ MULTQ3							0				Sol
TimeTravellingSale	SRM492-D2-1000							0				
March of the Peng	UVA 12125							0				Sol
PeopleYouMayKnow	SRM447-D1-500							0				Don't use DP. Check it later in editorial. Sol
The Game of 31	UVA 10578							0				Sol
Can you answer th	SPOJ GSS4							0				Sol
Area	TJU 1011							0				Sol
Volatile Kite	CF801-D2-D							0				Sol
Antifloyd	UVA 10987							0				Sol
Messenger	CF631-D2-D							0				
	SRM144-D1-500							0				
	SRM509-D1-500							0				
	CF280-D1-C							0				
	CF110-D2-D							0				
	CF163-D12-C							0				
	CF455-D1-B							0				
								0				
Infinite Maze	CF197-D2-D							0				
Jeff and Furik	CF352-D2-D							0				Sol
Sagheer and Kind	CF812-D2-D							0				Sol
Dispute	CF242-D2-D							0				
Remainders Game	CF688-D2-D							0				
	CF1075-D2-D							0				
	CF1033-D12-D							0				
	CF442-D1-B							0				
	CF1025-D2-D							0				
	CF1072-D2-D							0				

ff	Problem Code	Status	Submit Count	Reading Time(m)	Thinking Time(m)	Coding Time(m)	Debug Time(m)	Total Time(m)	Problem Level /10	By yourself?	Category	1-2 line Comments about your approach is interesting?	Mostafa Category	Category Code	Level	Quality
	AC Averages =>	0	0	0	0	0	0	0	0	0	0	0				
	CF101294-GYM-I							0				Sol	dp, dp_ranges	32	4.5	p1
	SRM441-D1-250							0					dp, dp_ranges, [consecutive ranges, cyclic p	32	5	p2
	SRM536-D2-1000							0					dp, dp_ranges, [consecutive ranges]	32	5	p1
MessageMess	SRM149-D1-500							0					dp, dp_ranges, impl, [consecutive ranges]	32	5	
	SRM555-D2-1000							0					dp, dp_ranges, [consecutive ranges]	32	5	
	SRM558-D1-250							0					dp, dp_ranges, [consecutive ranges] or bf	32	5.5	p2
Exploring Pyramid	UVA 1362							0				Video Solution - Eng Ayman Salah	dp, dp_ranges	32	5.5	
Brackets sequence	UVA 1626							0				Sol	dp, dp_ranges	32	5.5	
	SRM509-D1-500							0					dp, dp_ranges, floyd, [cases]	32	6	p4
	UVA 507							0					dp, dp_subrectangle, 1d, [more direct uva	36	3	
	UVA 10667							0					dp, dp_subrectangle, 2d	36	3	
Big Maximum Sum	CF75-D2-D							0					dp, dp_subrectangle, 2d, [actually greedy v	36	5	p2
	SPOJ FISHES							0				Sol	dp, dp_subrectangle, 2d, observations, dot	36	5.5	p3
Reberland Linguist	CF667-D2-C							0					dp, dp_table	37	4.5	p3
Red-Green Tower	CF478-D2-D							0					dp, dp_table, dp_roll	37	5	p3
Cunning Genie	CF418-D1-B							0					dp, dp_table, dp_roll, dp_bitmasks, sortings	37	5.5	p4
	ZOJ 3305							0				Sol	dp, dp_table or dp_bitmasks, all submasks	37	5.5	p4
An overnight danc	CF814-D2-D							0				Video Solution - Solver to be (Java)	dp, dp_trees, geometry or greedy	38	5	p3
	CF161-D12-D							0				Reading: DP on Trees	dp, dp_trees or dsu-on-trees	38	5	p2
Vertex Cover	SPOJ PT07X							0				Sol	dp, dp_trees	38	5	
	CF337-D2-D							0				Sol	dp, dp_trees or diameter like, [tricky to gue	38	5.5	p4
Chloe and please	CF743-D2-D							0					dp, dp_trees	38	5.5	p2
	Timus 1362							0				Sol	dp, dp_trees or greedy	38	5.5	p2
	UVA 1218							0				Sol	dp, dp_trees, [vertex cover related]	38	5.75	p2
Playing Cubes	CF257-D2-B							0					game theory, greedy	41	2.5	
Euclid's Game	UVA 10368							0				Video Solution - Eng Moaz Rashad	game theory, gcd, dfs or pattern, [why each	41	3.5	p2
	CF1220-D12-C							0					game theory, adhoc	41	3.5	p2
Alice and Bob	CF347-D2-C							0				Video Solution - Eng Mohamed Nasser	game theory, gcd	41	4	p1
Win or Freeze	CF151-D2-C							0				Video Solution - Dr Mostafa Saad	game theory, divisors, greedy	41	4	p1
Brownie Points	UVA 10865							0				Video Solution - Eng Magdy Hasan	geometry	45	2	p1
	SRM436-D2-500							0					geometry, [slopes comparison]	45	3	p1
Points in Figures:	UVA 476							0					geometry	45	3	
Watering Flowers	CF617-D2-C							0					geometry, bf	45	3	
Pouring Rain	CF667-D2-A							0					geometry, physics	45	3	
Fourth Point !!	UVA 10242							0				Video Solution - Eng Magdy Hasan	geometry, vectors addition	45	3	
Captain Marmot	CF474-D2-C							0				Video Solution - Dr Mostafa Saad	geometry, check square, point rotation, bf	45	3.5	p2
Overlapping Recta	UVA 460							0				Video Solution - Eng Muntaser Abukadeja	geometry	45	3.5	
Xrange's Pancake	HACKR xrange-and-piz							0				Sol	geometry, adhoc	45	4	p2
	HACKR a-circle-and-a-							0					geometry, ccw, parametric equ, in circle	45	4	p2
	SPOJ FACENEMY							0				Sol	geometry, angles, precision	45	4	p1
k-Multiple Free Se	CF275-D2-C							0					geometry	45	4	
Gerald's Hexagon	CF560-D2-C							0					geometry	45	4	
View Angle	CF257-D2-C							0				Editorial - Eng Ahmed Osama	geometry, angles	45	4	
Watchmen	CF651-D2-C							0					geometry, datastructures	45	4	
Bicycle Race	CF659-D2-D							0					geometry, impl, [very nice, o(1) and o(n) sol	45	4.5	p3
Pyramids	SPOJ PIR							0				Sol	geometry, formula or matrix determinant	45	4.5	p1
Pythagorean Triplet	CF707-D2-C							0					geometry, triangles, formula	45	4.5	p2
	SPOJ BILLIARD							0				Sol	geometry, angles, physics	45	5	
Cupboard and Ball	CF342-D2-C							0					geometry	45	5	p3
	CF1064-D2-E							0					geometry, binary search, interactive	45	5	p3
	CF961-D12-D							0					geometry	45	5	p2
	CF101917-D12-E							0					geometry, [lpl scared in contest, but easy]	45	5	p2
	CF552-D2-D							0					geometry, bf, counting, treemaps	45	5	p2
	CF1016-D2-E							0					geometry, binary search	45	5	p2
	CF1058-D2-D							0					geometry, triangles, number theory	45	5	p2
	UVA 1342							0				Sol	geometry, plane graph	45	5	
	CF101864-GYM-L							0				Sol	geometry, binary search or bf, greedy	45	5.5	p3
	CF80-D2-D							0					geometry, probability or algebra	45	5.5	p2
	UVA 11648							0				Sol	geometry, trapezoid formula, binary search	45	6	p2
	UVA 1333							0				Sol - Text/Background Clarification	geometry, triangles, angles, parallelogram l	45	6	p1
Hit Ball	CF203-D2-D							0					geometry, 3d, impl, math, [physics, kinemat	46	5	p2
	UVA 453							0				Learn Handling Precisions	geometry, circles, [direct circle intersection,	47	2	
Wifi Access	UVA 12748							0				Sol	geometry, circles, distances	47	2	
Rings and Glue	UVA 10301							0				Sol	geometry, circles, dsu	47	3	p1
Square Pegs And	UVA 356							0				Sol to read	geometry, circles	47	3	
The Circumferenc	UVA 438							0				Sol	geometry, circles	47	3	
Points in Figures:	UVA 477							0				Sol	geometry, circles	47	3.5	
Special Olympics	CF199-D2-B							0					geometry, circles, impl	47	4	
Biathlon	CF84-D2-C							0					geometry, circles, impl	47	4	
Packing polygons	UVA 10005							0				Sol	geometry, circles, polygon, [polygon inside p	47	5	p4
	SRM473-D1-500							0					geometry, circles, triangles, thales' theorem	47	5	p3
	SPOJ ALIENS							0				Sol - Practice on min enclosing circle	geometry, circles, min enclosing circle, [=sp	47	5	p2
	CF1059-D2-D							0					geometry, circles, binary search	47	5.25	p3
	HACKR house-location							0				Sol	geometry, circles, algebra, impl	47	5.5	p3
	UVA 10180							0				Sol	geometry, circles, tangents, point on segme	47	5.5	p2
Railway	UVA 10263							0				Sol to read	geometry, lines, distances, [=uva 460]	48	3	p3
Lining Up	UVA 270							0				Video Solution - Eng Mohamed Nasser, Don't Co	geometry, lines, line up	48	3	p3
Campus Roads	UVA 11473							0				Sol	geometry, lines, distances, impl	48	3	p2
Polyline	CF617-D2-D							0					geometry, lines, impl	48	3	
Jack Straws	UVA 273							0				Sol	geometry, lines, intersection, shortest path	48	3	
Isolated Segments	UVA 11343							0				Sol	geometry, lines, intersections	48	3	
Intersecting Lines	UVA 378							0					geometry, lines	48	3.5	
	SRM373-D2-1000							0				Sol	geometry, lines, lines intersection, rectangle	48	4	
Intersecting Line S	UVA 866							0				Sol	geometry, lines, intersections	48	4	
	SRM368-D1-500							0				Sol	geometry, lines, polyline intersection, bf, na	48	4	
Gleaming the Cub	UVA 737							0				Sol	geometry, lines, intersections	48	4	
Water Falls	UVA 833							0				Sol	geometry, lines, distances, adhoc	48	4	p3
How Many Points	UVA 10790							0				Sol	geometry, lines, intersections, counting, for	48	4	p1
River Crossing	UVA 10514							0				Sol	geometry, lines, distances, floyd	48	5	
Best Triangulation	SRM278-D2-500							0					geometry, polygon, area, [just triangle area	49	2	
Triangle	CF408-D2-C							0					geometry, polygon	49	4	
	UVA 11665							0				Sol	geometry, polygon, pip, polygons intersect	49	4	
	TIMUS 1599							0				Sol	geometry, polygon, pip, winding numbers, [l	49	4.5	p1
	UVA 881							0				Sol	geometry, polygon, pip, polygons inside pol	49	4.5	
	CF340-D2-B							0					geometry, polygon, bf	49	5	p2
Volatile Kite	CF801-D2-D							0				Sol	geometry, polygon, binary search	49	5	p2
Polygons	UVA 137							0				Sol	geometry, polygon, pip, intersections or cor	49	5.5	p3
Area	TJU 1011							0				Sol	geometry, polygon, pick's theorem	52	4.5	p1
Trees on My Islan	UVA 10088							0					geometry, polygon, pick's theorem, gcd	52	5	
	LIVEARCHIVE 2831							0				Use polygon cut	geometry, polygon, polygon cut	53	4	
Video Surveillance	UVA 588							0				Use polygon cut	geometry, polygon, polygon cut or adhoc	53	6	p5
The Skyline Proble	UVA 105							0					greedy, geometry	55	3	
Marcus	UVA 10452							0				Video Solution - Eng Ayman Salah	graph	55	3	
Trees on the level	UVA 122							0				Video Solution - SolverToBe (Java)	graph, trees	55	3	
PT07Z	SPOJ PT07Z							0				Sol	graph, tree diameter	55	3	
Roads in the North	UVA 10308							0				Sol	graph, tree diameter	55	3	
	CF1068-D2-C							0					graph, adhoc	55	4	p2

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	AC Averages ==>	0	0	0	0	0	0	0	0	0	0	0				
Eternal Victory	CF61-D2-D							0					graph, greedy	55	4	p2
Is It A Tree?	UVA 615							0					graph, trees	55	4	p1
Mahmoud and Elh	CF959-D2-C							0				Video Solution - Eng Mohamed Salah	graph, trees, constructive	55	4	
Central Post Office	UVA 12379							0				Sol	graph, tree diameter	55	4	
The Tree Root	UVA 10459							0				Sol	graph, tree diameter	55	4.5	p3
Xor-tree	CF430-D2-C							0					graph, bf	55	5	
Renting Bikes	CF363-D2-D							0					graph, cycle, greedy	55	5	
Regular Bridge	CF550-D2-D							0					graph, prove using e.g. soc	55	5	
	CF486-D2-D							0					graph, trees, dfs, prefix sum or dp_trees	55	5.5	p5
Cycles	CF233-D2-C							0					graph, cycle	55	5.5	p3
	CF459-D2-E							0					graph, dp, sortings	55	5.5	p3
	CF1060-D12-D							0					graph, greedy	55	5.5	p3
	UVA 10982							0				Sol	graph, greedy, [close to max cut]	55	5.5	p3
	CF592-D2-D							0					graph, tree diameter	55	5.5	p3
BITMAP - Bitmap	SPOJ BITMAP							0					graph, bfs, multiscr, multidest	57	3	p3
Pouring water	SPOJ POUR1							0				Video Solution - Eng Moaz Rashad	graph, bfs	57	3	
Jugs	UVA 571							0				Video Solution - Dr Mostafa Saad	graph, bfs	57	4	p1
Tic-Tac-Toe (I)	SPOJ TOE1							0				Video Solution - Eng Ayman Salah	graph, bfs	57	4	
Tic-Tac-Toe (II)	SPOJ TOE2							0				Video Solution - Eng Essam AlNaggar	graph, bfs	57	4	
Knight Moves	UVA 439							0				Video Solution - Eng Magdy Hasan	graph, bfs, chess or dfs	57	4	
King's Path	CF242-D2-C							0				Video Solution - Dr Mostafa Saad	graph, bfs	57	4.5	
Theseus and labyr	CF676-D2-D							0					graph, bfs, impl	57	4.5	p2
Wandering Queen	SPOJ QUEEN							0				Sol to read	graph, bfs	57	4.5	p1
Restore Graph	CF404-D2-C							0					graph, bfs	57	4.5	
Key Task	SPOJ CERC07K							0					graph, bfs, bitmask	57	4.5	
Cleaning Robot	SPOJ CLEANRBT							0					graph, bfs, bitmask or bfs preprocess then	57	4.5	
	UVA 10888							0					graph, bfs, dp or weighted matching	57	5	p3
Text Editor	CF253-D2-C							0					graph, bfs or greedy, [search in 2d grid]	57	5	p2
Tobo or not Tobo	SPOJ ANARC08A							0				Sol	graph, bfs, trie, hashing or meet in middle	57	5	
	CF1005-D3-F							0					graph, bfs	57	5.25	p2
	TIMUS 1498							0					graph, bfs, [chess, tricky cases]	57	5.5	p2
	UVA 11573							0				Learn 0/1 BFS	graph, bfs, 0/1 bfs, [-spo] kathi]	57	5.5	p2
	CF787-D2-C							0					graph, bfs, cyclic games	57	5.5	p1
	CF811-D2-D							0					graph, bfs, interactive	57	6	p2
	UVA 10461							0					graph, dfs, [finish computation times]	60	3	p1
Roads in Berland	CF25-D2-C							0					graph, dfs	60	4	p2
Party	CF116-D2-C							0					graph, dfs	60	4	p1
Forming Teams	CF216-D2-B							0				Video Solution - Dr Mostafa Saad	graph, dfs	60	4	
Block Tower	CF327-D2-D							0					graph, dfs	60	4	
Soldier and Cards	CF546-D2-C							0					graph, dfs	60	4	
Kefa and Park	CF580-D2-C							0				Video Solution - Solver to be (Java)	graph, dfs	60	4	p2
Maze	CF378-D2-C							0					graph, dfs, [reverse thinking]	60	4.5	p1
Exchange Rates	UVA 10113							0					graph, dfs, impl	60	4.5	p1
Ice Cave	CF540-D2-C							0					graph, dfs	60	4.5	
Ordering	UVA 872							0					graph, dfs	60	4.5	
Directed Roads	CF711-D2-D							0					graph, dfs, combinatorics, formula	60	5	p3
	SPOJ BIA							0				Sol	graph, dfs or directed articulation points alg	60	5	p2
Choosing Capital f	CF219-D2-D							0					graph, dfs or dp_trees	60	5	
	CF1075-D2-D							0					graph, dfs, interactive	60	5.5	p3
Infinite Maze	CF197-D2-D							0					graph, dfs	60	5.5	
Cycle in Graph	CF263-D2-D							0					graph, dfs	60	5.5	
T-decomposition	CF237-D2-D							0					graph, dfs, greedy	60	5.5	
Robbery	UVA 707							0				Sol	graph, dfs or dp	60	5.75	
Persistent Bookca	CF707-D2-D							0				Sol	graph, dfs, bitset or persistent segment tree	60	6	p3
Moodular Arithmet	CF604-D2-D							0				Sol	graph, dfs, fermat, [rearrangement property]	60	6	p2
The Seasonal War	UVA 352							0				Video Solution - Eng Mohamed Nasser	graph, dfs, flood-fill	61	2	
Battleships	UVA 11953							0				Video Solution - Eng Aya Elymany	graph, dfs, flood-fill	61	3.5	
Maze Exploration	UVA 784							0				Video Solution - Eng Mahmoud Adel	graph, dfs, flood-fill	61	3.5	
Continents	UVA 11094							0				Video Solution - Eng Ayman Salah	graph, dfs, flood-fill	61	4	
	SRM297-D1-500							0					graph, dfs, flood-fill or bfs, bf	61	5	p3
Equivalent Strings	CF560-D2-D							0				Sol to learn	graph, dfs, isomorphism or d&c, hashing	62	4	p2
Subway tree syste	LIVEARCHIVE 2935							0				Sol	graph, dfs, isomorphism, canonical form or	62	4.5	p4
Hierarchy	SPOJ MAKETREE							0				Video Solution - Eng Yahia Ashraf	graph, dfs, topological sort	63	2	
Ordering Tasks	UVA 10305							0				Video Solution - Eng Yahia Ashraf	graph, dfs, topological sort	63	3	
	SRM419-D2-1000							0					graph, dfs, topological sort, cycles	63	4	
Spreadsheet	UVA 196							0					graph, dfs, topological sort or dp	63	4	p3
Rankings	UVA 12263							0				Sol	graph, dfs, topological sort	63	4	p2
Pick up sticks	UVA 11686							0				Sol	graph, dfs, topological sort, detect cycles	63	4	
	SRM550-D2-1000							0					graph, dfs, topological sort	63	5	p3
Robot Rapping Re	CF645-D12-D							0					graph, dfs, topological sort, binary search	63	5	p3
Gifts by the List	CF681-D2-D							0					graph, dfs, topological sort, impl	63	5	p2
Sagheer and Kind	CF812-D2-D							0				Sol	graph, dfs, topological sort or euler, [https://	63	6	p4
Shopping	SPOJ SHOP							0					graph, dfs, topological sort or euler, [https://	63	6	p4
Sending email	UVA 10986							0					graph, dijkstra	64	3	
MELE3	SPOJ MELE3							0				Sol	graph, dijkstra	64	3	
Roads	SPOJ ROADS							0				Sol	graph, dijkstra or dp	64	4.5	p3
Lift Hopping	UVA 10801							0					graph, dijkstra	64	4.5	
	UVA 10740							0				Sol	graph, dijkstra, kth sp. [k <= 10]	64	5	p3
Volleyball	CF96-D2-D							0					graph, dijkstra, 2 dijkstra	64	5	p3
	UVA 12047							0				Sol	graph, dijkstra	64	5.5	p3
	UVA 10342							0				Sol - read the statement clarification	graph, dijkstra, kth sp (k=2) or floyd	64	5.5	p3
Hotel booking	UVA 11635							0				Sol	graph, dijkstra	64	5.5	
IP-TV	UVA 1174							0					graph, dsu	65	2	
Count the Faces.	UVA 10178							0				Read first Euler Formula	graph, dsu or dfs, cycles	65	4	p2
Learning Language	CF278-D2-C							0					graph, dsu	65	4	
Virtual Friends	UVA 11503							0				Video Solution - Eng Moaz Rashad	graph, dsu	65	4	
Almost Union-Find	UVA 11987							0				Sol	graph, dsu	65	4.5	p3
Cthulhu	CF104-D2-C							0					graph, dsu	65	4.5	
The Child and Zoo	CF437-D2-D							0					graph, dsu	65	5	
Mahmoud and a D	CF766-D2-D							0				Video Solution - Solver to be (Java)	graph, dsu, [offline processing]	65	5	p3
	CF1012-D1-B							0					graph, dsu	65	5.25	p2
	UVA 12128							0					graph, dsu, dijkstra like or binary search, bf	65	5.5	p2
Connected Compco	CF292-D12-D							0					graph, dsu	65	5.75	p3
Trip Routing	UVA 186							0				Sol	graph, floyd, path print	68	4	p3
Numbering Paths	UVA 125							0				Sol	graph, floyd, paths counting	68	4.5	p5
Frogger	UVA 534							0				Sol	graph, floyd, minimax or dsu	68	4.5	p4
Travel in Desert	UVA 10816							0				Sol	graph, floyd, binary search	68	4.5	
Identifying Concur	UVA 334							0					graph, floyd	68	4.5	
Greg and Graph	CF296-D2-D							0					graph, floyd	68	5	p2
Dima and Bacteria	CF400-D2-D							0					graph, floyd, dfs	68	5	p2
AlgoRace	CF189-D2-D							0				Sol	graph, floyd	68	5.25	p4
Antifloyd	UVA 10987							0				Sol	graph, floyd, antifloyd	68	5.5	p4
Unique World	UVA 10448							0				Video Solution - Dr Mostafa Saad	graph, floyd, dp	68	5.5	p2
Arbitrage	UVA 104							0				Sol	graph, floyd	68	6.25	p2
Potholers	SPOJ POTTHOLE							0				Sol	graph, max-flow	71	3	
Power Transmissk	UVA 10330							0				Sol	graph, max-flow, vertex constraints	71	4	
The Problem with	UVA 10092							0					graph, max-flow, [direct bipartite is slow]	71	4.5	p3

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	AC Averages =>	0	0	0	0	0	0	0	0	0	0	0				
Tanya and Postcards	CF518-D2-B							0					impl	86	2	
Mike and Fun	CF548-D2-B							0					impl	86	2	
Covered Path	CF534-D2-B							0					impl	86	2	
Print Check	CF631-D2-B							0					impl	86	2	
Lucky Mask	CF146-D2-B							0					impl	86	2	
Special Offer! Super	CF219-D2-B							0					impl	86	2	p2
Non-square Equations	CF233-D2-B							0					impl	86	2	
Flag Day	CF357-D2-B							0					impl	86	2	
Sereja and Mirrors	CF426-D2-B							0					impl	86	2	
Little Pony and So	CF454-D2-B							0					impl	86	2	
MUH and Importers	CF471-D2-B							0					impl	86	2	
Gena's Code	CF614-D2-B							0					impl	86	2	
Opposites Attract	CF131-D2-B							0					impl	86	2	
Little Pigs and Wolves	CF116-D2-B							0					impl	86	2	
Cosmic Tables	CF222-D2-B							0					impl	86	2	
Prime Matrix	CF271-D2-B							0					impl	86	2	
Wet Shark and Bits	CF621-D2-B							0				Video Solution - Eng Mahmoud Mabrok	impl	86	2	
	CF1030-D12-B							0					impl, math	86	2	
Facetook Priority Queue	CF75-D2-B							0					impl, sorting	86	2	
	CF490-D2-B							0					graph, constructive, adhoc	86	3	p2
Hanoi Tower	TIMUS 1054							0			Sol		impl, recursion, tower of hanoi	86	3	p2
Treasure	CF495-D2-C							0					impl	86	4	
Game	CF68-D2-C							0					impl	86	4	
Accordian Patience	UVA 127							0			Video Solution - Eng Moaz Rashad		impl	86	4	p1
Beautiful Sets of Pairs	CF268-D2-C							0					impl, constructive	86	4	
Appleman and Tree	CF462-D2-C							0			Sol		impl, sorting, huffman coding	86	4	
Three Logos	CF581-D2-D							0					impl	86	4.5	p2
Guess Your Way Out	CF507-D2-C							0			Video Solution - Dr Mostafa Saad		impl, math	86	4.5	p2
	CF1042-D12-D							0					impl or segment tree or bit	86	5	p2
Mafia	CF349-D2-C							0					impl, math	86	5	p2
Unusual Product	CF405-D2-C							0					impl, math, [symbolic thinking]	86	5	p1
	CF101187-GYM-F							0			Sol		impl	86	5.25	p2
Special Grid	CF435-D2-D							0					impl, greedy	86	5.5	p2
Theatre Square	CF1-D12-A							0			Video Solution - Solver to be (Java)		math	87	1.5	
	CF1204-D2-A							0			Video Solution - Dr Mostafa Saad		math, log, binary, pattern observation	87	2	p2
Balanced Rating Contest	CF1237-D12-A							0					math, analysis	87	2	p3
The Drunk Jailer	LIVEARCHIVE 2557							0			Find a formula		math or bf	87	2	p1
Product	UVA 10106							0			Video Solution - Eng Youssef El Ghareeb_Don't		math	87	2	
To Carry or not to	UVA 10468							0			Sol		math	87	2	
Adding Reversed Integers	UVA 713							0			Don't use big integer class. Write simple array		math	87	2	
Dreamoon and WiFi	CF476-D2-B							0			Video Solution - Eng Mohamed Adel		math, combinatorics, bitmasks	87	2	p3
	CF1051-D2-B							0					math	87	2.5	
Escape	CF148-D2-B							0					math	87	2.5	
Restoring Painting	CF675-D2-B							0					math	87	2.5	
Caisa and Pylons	CF463-D2-B							0			Video Solution - Eng Muntaser Abukadeja		math, impl	87	2.5	
T-primes	CF230-D2-B							0					math, number theory	87	2.5	
	CODECHEF_GCDMOD							0			Sol uses __int128 to avoid overflow		math, __int128	87	3	p3
Polycarpus' Dilemma	CF534-D2-C							0			Sol		math, greedy, careful impl	87	3	p3
	CF1059-D2-C							0					math, constructive	87	4	p3
Number Sequence	UVA 10706							0					math	87	4	p2
Divisible by Seven	CF376-D2-C							0					math, number theory	87	4	p2
Fractions Again?!	UVA 10976							0			Sol to read		math, number theory	87	4	p1
Plant	CF186-D2-C							0					math	87	4	
Magic Formulas	CF424-D2-C							0					math	87	4	
Duff in Love	CF588-D2-B							0					math	87	4	
Light, more light	UVA 10110							0			Video Solution - Eng Amr Saud		math	87	4	
Power of Cryptography	UVA 113							0			Sol to read		math, log, [double limits]	87	4	p3
Round Table Knight	CF71-D2-C							0					math or dp	87	4	
Lucky Permutation	CF304-D2-C							0					math, constructive	87	4	
Vasya and Petya's Game	CF577-D2-C							0					math, impl	87	4	p2
	CF1239-D1-A							0					math, pattern	87	4.25	p3
The ? 1 ? 2 ? ... ?	UVA 10025							0					math or binary search	87	4.5	p2
Secrets	CF334-D2-C							0					math	87	4.5	
The Meaningless Contest	CF834-D2-C							0			Video Solution - Solver to be (Java)		math	87	4.5	
Find Maximum	CF353-D2-C							0					math, bits	87	4.5	
Plus and Square Root	CF716-D2-C							0					math, constructive	87	4.5	
Bear and Prime 10	CF680-D2-C							0					math, constructive, interactive	87	4.5	
	CF1040-D2-D							0					math, randomization, binary search, interaction	87	5	p4
Count Good Substrings	CF451-D2-D							0					math, adhoc, palindromes, [short code]	87	5	p3
Tavas and Karafs	CF535-D2-C							0					math, binary search	87	5	p2
As Fast As Possible	CF701-D2-D							0					math, binary search, precision	87	5	p2
	CF955-D2-C							0					math, number theory	87	5	p2
	CF45-D12-D							0					math, randomization	87	5	p2
Ciel and Robot	CF322-D2-C							0					math, impl, [cases]	87	5	p1
Crazy Town	CF499-D2-C							0			Video Solution - Dr Mostafa Saad		math, number theory, greedy	87	5	p1
About Bacteria	CF199-D2-C							0					math	87	5	
DNA Alignment	CF520-D2-C							0					math	87	5	
Predict Outcome of Game	CF451-D2-C							0					math, equations, impl	87	5	p2
Analyzing Polyline	CF195-D2-D							0					math, sortings	87	5	
	CF1016-D12-D							0					math, xor, bitwise, constructive	87	5.25	p3
Quantity of Strings	CF151-D2-D							0					math, repeated squaring, graph	87	5.5	p4
How many trees?	CF9-D2-D							0					math or dp_tree	87	5.5	p2
The Errant Physicist	UVA 128							0			Sol		math	87	5.5	
Software CRC	UVA 128							0			Video Solution - Eng Moaz Rashad		math	87	5.5	
Jeff and Furk	CF352-D2-D							0			Sol		math or dp_expectation	87	6	p3
Magical Array	CF84-D2-B							0					math, combinatorics	89	2.5	
Chocolate	CF617-D2-B							0			Video Solution - Eng Yahia Ashraf		math, combinatorics	89	2.5	
Colorful Field	CF79-D12-B							0			Video Solution - Solver to be (Java)		math, mod	89	2.5	p3
The World is a Theatrical Performance	CF131-D2-C							0			Video Solution - Eng Youssef Ali		math, combinatorics	89	4	
Pocket Book	CF152-D2-C							0					math, combinatorics	89	4	
Black and white photograph	UVA 11231							0			Video Solution - Eng Amr Saud		math, combinatorics, counting	89	4	
	CF758-D2-C							0					math, combinatorics	89	5	p3
	CF459-D2-C							0					math, combinatorics, constructive	89	5	p3
	HACKER_ajourney							0					math, combinatorics, first/last k digits 2^n, [	89	5	p3
Shaass and Lights	CF294-D2-C							0			Video Solution - Dr Mostafa Saad		math, combinatorics	89	5.5	p4
	CF869-D2-C							0					math, combinatorics or dp_counting	89	5.5	p3
Tourist Problem	CF340-D2-C							0					math, combinatorics, impl	89	5.5	p1
Fox Dividing Cheesecake	CF371-D2-B							0			Video Solution - Eng Abanob Ashraf		math, factorial	94	2.5	
Permalex	UVA 153							0			Sol		math, factorial, permutations, duplicates, factorization	94	4.5	p3
Prime Factors	UVA 583							0					math, factorization	95	2	
Easy Number Characterization	CF236-D2-B							0			Video Solution - Eng Yahia Ashraf		math, factorization	95	3	
Mr. Azad and his Friends	UVA 10490							0			Sol to read		math, factorization	95	3	
Prime Land	UVA 516							0					math, factorization	95	3	
Perfect P-th Power	UVA 10622							0			Video Solution - Eng Moaz Rashad		math, factorization	95	4	p1
Factovisors	UVA 10139							0			Sol to read		math, factorization, primes, [factorize x!]	95	4	
	CF1047-D2-C							0					math, factorization	95	4.5	p3
DDF	UVA 547							0					math, factorization, divisors sum, multiview	95	4.5	

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	AC Averages => UVA 10174	0	0	0	0	0	0	0	0	0	0	0	math, factorization, case analysis	95	5	
	UVA 11347							0					math, factorization, divisors sum	95	5	
Multifactorials	CF1033-D12-D							0					math, factorization	95	5.5	p3
Remainders Game	CF688-D2-D							0					math, factorization, gcd, lcm, observations	95	6	p4
Primitive Root	SPOJ PROOT							0				Sol	math, factorization, primitive roots	95	6.25	p4
	UVA 12869							0				Sol	math, formula	98	5	p2
Combinations	UVA 369							0					math, gcd, comb formula	99	2	
Pi	UVA 412							0				Video Solution - Eng Mohamed Adel	math, gcd	99	3	
Trains	CF88-D2-C							0				Video Solution - Solver to be (Java)	math, gcd or adhoc	99	4	
Mint	UVA 10717							0				Sol	math, gcd, lcm	99	4	
The Big Race	CF592-D2-C							0					math, gcd, lcm, [overflow]	99	4.5	p3
LCM Cardinality	UVA 10892							0					math, gcd, lcm	99	4.5	
Rational Resistance	CF344-D2-C							0					math, gcd	99	5	p3
LCM Challenge	CF236-D2-C							0					math, gcd, lcm	99	5	
	CF1010-D1-C							0					math, gcd, mod, number theory	99	5.5	p1
	AICoder026-AGC-B							0				Sol	math, gcd, cases	99	6	p3
	SPOJ EASYMATH							0				Sol	math, inclusion-exclusion, lcm	101	3	
Hamburgers	CF371-D2-C							0					math, inclusion-exclusion, binary search	101	3	
The Lottery	UVA 10325							0				Video Solution - Eng Amr Bahaa [4]	math, inclusion-exclusion, gcd, overflow	101	4	
	CF101933-GYM-K							0				Sol	math, inclusion-exclusion	101	4	
	CF372-D1-B							0					math, inclusion-exclusion	101	5.75	p2
	SPOJ MSKYCODE							0				Sol	math, inclusion-exclusion	101	6	p3
	CF101992-GYM-D							0				Sol	math, inclusion-exclusion	101	6	p3
Equation	UVA 727							0					math, infix to postfix	102	4	
Farm	TIMUS 1349							0				Know Fermat's Last Theorem (ignore proof)	math, math_adhoc, fermat last theorem	104	2	p2
Summation of Poly	UVA 10302							0					math, math_adhoc, polynomials	104	2	
	HACKR lower-3-colorin							0				Learn Fermat's little theorem	math, math_adhoc, fermat little theorem	104	3	p1
R U Kidding Mr. Fr	UVA 10509							0					math, math_adhoc, patterns	104	3.5	
Polly the Polynomi	UVA 498							0					math, math_adhoc, polynomials	104	3.5	
Jzzhu and Sequen	CF450-D2-B							0					math, matrix, matrix exponent	105	2.5	
Mirror, Mirror	UVA 466							0					math, matrix, rotate, reflect, impl	105	3	p1
Clear Symmetry	CF202-D2-C							0					math, matrix, bf	105	4	
End of Fun	SPOJ DCEPC12E							0					math, matrix	105	4.5	
Uniform Generator	UVA 408							0				Video Solution - Eng Yahia Ashraf	math, mod	109	3	
Be Efficient	UVA 11155							0					math, mod	109	5	
Quiz	CF337-D2-C							0					math, mod, pow, greedy	109	5.5	p3
Cows and Cars	UVA 10491							0				Revise Probability	math, probability, formula, fraction style	113	2	
What is the Probat	UVA 10056							0				Sol	math, probability	113	3	
	HACKR sherlock-and-p							0				Sol	math, probability, fractions style	113	3	
Probability(Given	UVA 11181							0				Sol	math, probability, conditional probability	113	4	p2
Another lottery	UVA 11628							0				Sol	math, probability, fraction style, gcd	113	4	
Mushroom Scientists	CF186-D2-D							0					math, probability or log, ternary search	113	5	p4
	CF101864-GYM-A							0				Sol	math, probability, combinatorics, math	113	5	p2
	SRM537-D2-1000							0					math, probability, graph, cycle	113	5	p2
Airplane	UVA 12461							0				Sol	math, probability, greedy	113	5	p1
Probability	Uva 11346							0				Sol	math, probability, integration	113	5.25	p3
	SRM285-D1-500							0					math, probability, bf or dp	113	5.5	
	CF26-D12-D							0				Sol - must read	math, probability, factorial, logarithm, combi	113	5.5	p3
	CF442-D1-B							0					math, probability, sorting	113	5.5	p3
	SRM352-D2-1000							0					math, probability, recursion, precision	113	5.5	
	CF513-D12-C							0				Sol	math, probability, bitmasks or dp, probability	113	6	p3
	UVA 557							0				Sol	math, probability, combinatorics	113	6	
	SPOJ FUNPROB							0				Sol	math, probability, formula	113	6	
	CF163-D12-C							0					math, probability	113	6.25	
	CF110-D2-D							0					math, probability, combinatorics	113	6.25	
God, Save me	UVA 10777							0				Sol	math, probability, expectation or dp, probab	114	4	
	SRM458-D2-500							0					math, probability, expectation, bitmasks	114	4	
	CF839-D2-C							0					math, probability, expectation, dfs	114	4	
	HACKR lazy-sorting							0				Revise Expected Value	math, probability, expectation, permutation	114	4	
Andrey and Probab	CF443-D2-D							0				Sol	math, probability, expectation, greedy or dp	114	4.5	p3
Wet Shark and Flo	CF621-D2-C							0					math, probability, expectation	114	4.5	
Little Pony and Ex	CF454-D2-C							0					math, probability, expectation, pattern	114	4.5	
	HACKR vertical-sticks							0					math, probability, expectation, linearity of ex	114	5	p3
	SRM577-D1-250							0					math, probability, expectation, linearity of ex	114	5	p3
	SRM470-D1-500							0					math, probability, expectation	114	5.5	p2
	CF500-D12-D							0					math, probability, expectation, dfs	114	5.5	p2
	CF280-D1-C							0					math, probability, expectation, dfs or dp	114	6	p3
Playlist	CF268-D2-E							0				Sol	math, probability, expectation, formula, gre	114	6	p3
Big Mod	UVA 374							0					math, repeated squaring, mod, direct	115	3	
Twin Primes	UVA 10394							0					math, sieve	117	3	
Factorial Factors	UVA 884							0					math, sieve, factorization	117	3.5	
Psycho	SPOJ PSYCHON							0					math, sieve, factorization, tricky big # test c	117	4	p2
Summation of Fou	UVA 10168 [5]							0				Video Solution - Eng Moaz Rashad [6]	math, sieve	117	4	
Primes or Palindro	CF569-D2-C							0					math, sieve, palindromes	117	4.5	p3
Divisibility of Facto	UVA 10484							0				Sol to read	math, sieve	117	4.5	p2
	LIVEARCHIVE 4008							0					math, sieve, [last non zero digit of permutat	117	5.5	p2
The New Rule in E	UVA 10742							0				Sol	math, sieve, binary search	117	5.5	
Sum-up the Prime	UVA 10419							0				Sol	math, sieve, dfs, dp	117	5.5	
Flying Saucer Seg	CF227-D2-C							0					math, summations	118	4.5	p3
Dreamoon and Su	CF476-D2-C							0				Video Solution - Dr Mostafa Saad	math, summations, [in my videos]	118	5	p3
	CF201-D1-B							0					math, summations, separate summations o	118	5	p2
Spongebob and S	CF599-D2-D							0					math, summations, bf, [overflow]	118	6	p2
Largest Rectangle	SPOJ HISTOGRA							0				Sol_Don't implement as adhoc/greedy/Pure ST	rmq, d&c or datastructure, [largest rectangl	122	4.5	p4
R2D2 and Droid A	CF514-D2-D							0				Use rmq	rmq, binary search or bit or two pointers	122	5	p3
Friends and Subst	CF689-D2-D							0					rmq, sparse table, binary search or datastru	122	5	p3
Pair of Numbers	CF359-D2-D							0				Sol	rmq, binary search, gcd, analysis or stack	122	5.5	p2
Square Subsets	CF448-D2-C							0					search, d&c, greedy	123	4.5	
Potentiometers	LIVEARCHIVE 2191							0					segment tree, [interval sum query]	125	2	p3
Interval Product	UVA 12532							0					segment tree or bit, [==tju 3440]	125	2	
Halt The War	SPOJ CDC12_H							0					segment tree	125	3.5	
Multiples of 3	SPOJ MULTQ3							0				Sol	segment tree, lazy propagation	125	4	p3
Horrible Queries	SPOJ HORRIBLE							0					segment tree, lazy propagation or bit	125	4	p1
Counting Primes	SPOJ CNTPRIME							0					segment tree, sieve	125	4	p1
Maximum Sum	SPOJ KGSS							0					segment tree, [max pair sum]	125	4.5	p3
A Famous City	SPOJ CITY2							0				Sol	segment tree or adhoc	125	4.5	p2
Help R2-D2!	SPOJ HELPR2D2							0					segment tree, impl	125	4.5	p2
Light Switching	SPOJ LITE							0					segment tree, lazy propagation, [edu]	125	4.5	p1
Circular RMQ	CF52-D12-C							0					segment tree, lazy propagation, circular	125	4.5	
Brackets	SPOJ BRCKTS							0				Sol	segment tree, [bracket balance, 2 values in	125	5	p3
Can you answer th	SPOJ GSS1							0				Sol	segment tree, [max sum, part of gss series	125	5	p3
RMQ with Shifts	UVA 12299							0				See sscanf and sprint usage	segment tree, rmq shift	125	5	p3
AND Rounds	SPOJ ANDROUND							0				Sol	segment tree	125	5	p2
Ahoy, Pirates!	UVA 11402							0				Sol	segment tree, lazy propagation or datastruc	125	5	p2
Present	CF460-D2-C							0					segment tree, lazy propagation, greedy or t	125	5	p2
Fence Obstacle Cr	PKU 2374							0				Sol	segment tree, dp or dp	125	5	p1
	CF61-D2-E							0					segment tree or wavelet tree, [boring, inver	125	5	p1
Can you answer th	SPOJ GSS3							0					segment tree, [max sum+updates, spoj gss	125	5.5	p5

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	AC Averages =>	0	0	0	0	0	0	0	0	0	0	0				
						62				63		Watch - Graph Theory - Tree Diameter and Isomorphism				
						114				64		Watch Video - Expected Value				
						122, 125				65		Watch - Data Structures - Segment Tree (2 vid)				
						38				66		Reading: DP on Trees				
						138				67		Watch - Two pointers technique				
						29				68		Watch - DP - Probability				
						11				69		Watch - DP - Masks (2 vid)				
						135				70		Watch - String Processing - Trie				
						36				71		Watch - DP - Sub-rectangle style				
						130				72		Watch - String Processing - KMP (2 vid)				
						23				73		Watch - DP - Games (2 vid)				
						49				74		Watch - Computational Geometry - Simple and Convex Polygons				
						49				75		Watch - Computational Geometry - Polygon Area - Centroid - Cut				
						49				76		Watch - Computational Geometry - Point in polygon				
						71,72,74				77		Watch - Graph Theory - Maximum Flow (2 vid)				
						77				78		Watch - Graph Theory - SCC (2 vid)				

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AC Averages =>		0	0	0	0	0	0	0	0	0	0	0				
		- Problems in this topic2 page don't exist in any other sheet (A, B, C, D, Topics1) - For juniors, you should ignore this page till finish the previous sheets FIRST - These topics represent 'most' of the other algorithms that seniors may learn - Problems per topic cover range of easy to medium level (hard Div2-D) - The purpose is to allow some of the trainees who want to expand in several algorithms before going deeper or whoever want to enhance a specific topic - 50-70% of the problems here will be of right level/quality. However, many of them are not evaluated by enough (or no) trainees. More future refinements may occur based on feedback - Help me refine this list. Consider these cases: - Problem that is so bad somehow - Problem of 'p3, p4, p5' quality but actually is a repeated idea or not interesting - Problem of 'empty, p1, p2' quality but actually is very interesting to solve - Problem of really wrong level (so easy/so hard) relative to my reported level If any of these issues, email me: mostafa.saad.fc@gmail.com Also, if a problem doesn't have an editorial and you could provide it = great!														
SRM629-D2-500													ternary search	10	2	
SPOJ KOPC12A													ternary search	10	3	
SPOJ TRICKTR													ternary search	10	3	
CF439-D2-D											Read the editorials		ternary search	10	3.5	
SRM287-D1-500													ternary search, math	10	4	
CF250-D2-D													ternary search, discrete, geometry	10	5	p5
UVA 10385													ternary search, tricky	10	5	p5
CF101102-GYM-I													ternary search	10	6	
CODECHEF_AMITNIT											Sol		meet in middle, bitmasks	11	3.5	
SPOJ SUBSUMS													meet in middle	11	4	p3
SPOJ ABCDEF													meet in middle	11	4	
SPOJ SUMFOUR													meet in middle, hashing, [solve SPOJ ABC]	11	4.5	
AtCoder026-AGC-C											Sol		meet in middle, bf, hasing	11	5	p3
CF490-D2-D													meet in middle, bfs	11	5	
CF525-D2-E													meet in middle, bitmasks, ternarymask	11	5.5	p1
CF1006-D3-F													meet in middle, dp	11	6	p3
CSA67-E													meet in middle, dp, bitmasks, [non standard	11	6.25	p5
CF585-D1-D													meet in middle, [strict tl]	11	6.25	p2
SPOJ MATSUM													bit, 2d bit	15	3	
SPOJ MSE06H													bit or merge sort, [SPOJ RATING]	15	4	p3
UVA 1428											Don't use segment tree or pure STL. Sol		bit or segment tree or datastructures	15	4	p2
UVA 11495													bit or merge sort, game, [count inversion]	15	4	
SRM315-D2-1000													bit or segment tree, [jouseph problem]	15	4.5	p2
SRM424-D1-1000													bit	15	5	
SPOJ ORDERSET											Sol		bit or segment tree or bbst, grid compress	15	5.5	p3
UVA 12697											Sol-must read		bit or segment tree or solutions-bag	15	6	p4
CF501-D2-D													bit, factoradics, permutation or order-statist	15	6	p4
SPOJ BRICKS											Sol		bit, count inversion or merge sort, [CODEC	15	6	p3
UVA 1513											Sol		bit or segment tree, [standard]	15	6	p2
CF538-D12-F													bit or segment tree, math or adthock	15	6.25	p4
SRM488-D1-250													dp, dp_cyclic, dp_expectation or dp_depth	17	5	
SRM568-D2-1000											Sol		dp, dp_cyclic, expectation, [algebra handlin	17	5	
SRM334-D1-500													dp, dp_cyclic	17	6	
SRM318-D1-500													dp, dp_cyclic, expectation or bf, simulation	17	6	
CF102190-GYM-B											Sol		dp, dp_cyclic, dp_probability or dp_depth	17	6.25	p3
UVA 11755																

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AC Averages =>		0	0	0	0	0	0	0	0	0	0	0	geometry, sweep line, sets	44	6.25	
SPOJ WILD												Sol	geometry, 3d, ternary search	45	3.5	
HACKR spheres												Sol	geometry, 3d, cones, volumes, formula	45	4	
UVA 10297												Sol	geometry, 3d, great circle distance	45	4	
UVA 11817													geometry, 3d, sphere, floyd	45	4.5	
LIVEARCHIVE 2233													geometry, 3d, impl, math, [physics, kinemat	45	5	p3
CF203-D2-D													geometry, 3d, lines, distances, binary searc	45	5	p1
CF65-D12-C													geometry, 3d, binary search	45	5	
SPOJ BLCONE												Sol	geometry, 3d, [differentiation needed for prc	45	5.5	
UVA 11232												Sol	geometry, 3d, integration, simpson, [cylinde	45	6	
ZOJ 2369													geometry, 3d, sphere, bf, next_permutation	45	6	
LIVEARCHIVE 2474													graph, bellmanford	47	3	
UVA 10449													graph, bellmanford	47	3.5	
UVA 10557													graph, bellmanford	47	4	
UVA 558												Sol	graph, bellmanford, [-CODECHEF BESTP]	47	6	p4
CF101498-GYM-L												Sol	graph, bellmanford, Minimum Mean Weight	47	6.25	p4
UVA 11090													graph, euler tour	54	3.5	
TIMUS 1137													graph, euler tour, dijkstra	54	3.5	
UVA 117													graph, euler tour, is there euler cycle?	54	3.5	
UVA 10596													graph, euler tour, handshaking lemma, cc	54	5	p1
SRM268-D1-500												Prove your solution	graph, euler tour	54	5	
UVA 10054													graph, euler tour, combinatorics	54	5.25	p3
CF788-D1-B													graph, euler tour, [solve first SRM268-D1-5]	54	6	p3
SRM298-D1-500													graph, euler tour, math, [required a solid un	54	6.1	p4
CF789-D2-D													graph, euler tour	54	6.1	p3
CF1038-D2-E													graph, min-cost-max-flow, weighted bipartit	59	4.5	
SPOJ SCITIES													graph, min-cost-max-flow	59	5	
UVA 10594													graph, min-cost-max-flow	59	6	p4
TJU 2554													graph, min-cost-max-flow	59	6	
TC(ANGELDEMONG/													graph, min-cost-max-flow	59	6	
LIVEARCHIVE 2884													graph, scc, 2-sat or greedy	62	3	
SPOJ BUGLIFE												Sol	graph, scc, 2-sat, [simple dfs, Bipartite grap	62	4	
LIVEARCHIVE 4185												Sol	graph, scc, 2-sat	62	5	
CF228-D2-E													graph, scc, 2-sat, dsu, topological sort or ge	62	5	
CF776-D2-D													graph, scc, 2-sat	62	5.5	
CF469-D2-D													graph, scc, 2-sat or dsu or greedy	62	6	p4
CODECHEF ADAMTFC													graph, scc, 2-sat	62	6	p3
LIVEARCHIVE 5010												Sol	graph, scc, 2-sat	62	6	
PKU 2723												Sol	graph, scc, 2-sat	62	6	
UVA 1146												Sol	graph, scc, 2-sat	62	6	
SRM464-D1-500													graph, scc, 2-sat	62	6.25	
UVA 10199													graph, scc, articulation point	63	3	p3
SPOJ SUBMERGE													graph, scc, articulation point	63	3	
UVA 315													graph, scc, articulation point	63	3	
UVA 10765													graph, scc, articulation point or dsu, bf	63	4	
UVA 796													graph, scc, biconnected components	64	4	
UVA 610												Sol	graph, scc, biconnected components, [=CF	64	5.5	p3
SPOJ QTREE2												Sol	graph, lca, [in video, ~PKU 1986, LIVEAR]	65	4	
CF192-D2-E												Sol	graph, lca, [or with dp, binary lifting]	65	5	p3
SPOJ DISQUERY												Sol	graph, lca, dp, binary lifting, [educational]	65	5	p3
CF832-D2-D													graph, lca, math	65	5	p2
CF519-D2-E													graph, lca, lca on tree, rmq, impl	65	5.5	p3
TIMUS 1752												Sol	graph, lca, tree diameter, [==KATTIS tourist	65	5.75	p3
CF468-D2-E													graph, lca or dsu, offline queries	65	6.25	p3
CF587-D1-C													graph, lca, dp, binary lifting, impl or segmen	65	6.25	p3
CF863-D12-E													grid compress, prefix sum or segment tree	67	6	
UVA 870												Find grid-compress based idea	grid compress, rectangles, dfs, counting or i	67	6	
CF243-D1-C													grid compress, dfs or bfs	67	6.5	
UVA 308												Sol	grid compress, flood-fill	67	6.5	
UVA 306													math, cyclic permutation	84	3	
SRM572-D1-250													math, cyclic permutation or dfs	84	3	
CF986-D1-B													math, cyclic permutation	84	5	p2
SRM391-D1-500													math, cyclic permutation, stirling number of	84	5	p2
SRM379-D2-1000													math, cyclic permutation or backtrack, dsu	84	5	
FbiHrCup 18-R1-B													math, cyclic permutation, dfs, trees	84	5.5	p2
SPOJ LEONARDO												Sol	math, cyclic permutation, graph cycles, eve	84	5.75	p4
SRM280-D2-1000												See Editorial	math, cyclic permutation or backtrack, pruni	84	5.75	p3
SPOJ SEQ													math, matrix, matrix pow	89	3	
SPOJ FIBTWIST													math, matrix, matrix pow, [fib]	89	3	
TJU 2300													math, matrix, matrix pow	89	3.5	p3
UVA 10229												Sol	math, matrix, matrix pow, fib, or pattern [dir	89	4	p2
SPOJ SUMSUMS												Sol	math, matrix, matrix pow, [basic hint in vide	89	4.5	p1
CF222-D2-E													math, matrix, matrix pow or dp	89	4.5	
CF582-D1-B													math, matrix, matrix pow, matrix pow max, c	89	5	p5
CF621-D2-E												Sol	math, matrix, matrix pow or dp, d&c	89	5.5	p3
SPOJ PLHOP												Sol	math, matrix, matrix pow, graph, matrix pow	89	6	p4
UVA 11605												Sol	math, matrix, matrix pow, linearity of expect	89	6	p3
LIVEARCHIVE 4332												Sol	math, matrix, matrix pow, recurrence, gene	89	6	p3
CF821-D2-E													math, matrix, matrix pow, impl	89	6	p2
SPOJ XMAX												Sol	math, matrix, gaussian elimination, gauss-x	90	5.5	p5
CF1155-D12-E													math, matrix, gaussian elimination	90	5.5	p2
CODECHEF TREASU													math, matrix, gaussian elimination, bitset	90	6	p3
TIMUS 1042												Editorial (Google Sol)	math, matrix, gaussian elimination	90	6	
UVA 684												Sol	math, matrix, gaussian elimination	90	6	
LIVEARCHIVE 4305												Sol(no editorial)	math, matrix, gaussian elimination, determi	90	6	
UVA 10104													math, extended gcd	94	3	
UVA 10673													math, extended gcd or gcd, simple math	94	3	
CF100812-GYM-L												Sol	math, extended gcd, totient	94	5	p1
UVA 10090												Sol	math, extended gcd	94	5	
HACKR solve-equatio												Test cases are wrong. Case 1 4 1 82: judge ou	math, extended gcd, geometry	94	5.5	
SRM385-D2-1000													math, diophantine	95	4.5	
LightOJ 1306												Sol	math, diophantine, extended gcd	95	5.5	p2
CF100506-GYM-C												Sol	math, diophantine, extended gcd or number	95	6	
UVA 11768												Sol	math, diophantine, extended gcd, lattice poi	95	6.25	
LIVEARCHIVE 5990												Sol	math, mod inv	96	4	
HACKR game-of-thro													math, mod inv, factorial inv or permutation	96	4	
CF327-D2-C													math, mod inv	96	5	
HACKR choose-and-c													math, mod inv, combinatorics, binomial coe	96	5	
SRM467-D1-500												Sol	math, mod inv, fermat's little theorem	96	5	
LIVEARCHIVE 4506													math, mod inv, dp, combinations	96	5.5	
SRM735-D1-500												Sol	math, mod inv, [easier version Timus 1132]	96	6	p3
CF146-D2-E													math, mod inv, factorial, dp_counting	96	6	p3
SPOJ DIVEQL												Sol	math, mod inv, gcd	96	6	p3
SPOJ KOPC12B												Sol - See CF post	math, mod inv, combinations, pattern	96	6	p2
CF816-D2-D													math, mod inv, combinatorics	96	6	p1
CF689-D2-E													math, mod inv, combinatorics, impl	96	6	
CF785-D2-D												Read this solution after trying	math, mod inv, summations, combinatorics	96	6.25	p5
CF521-D1-C												SolToMe	math, mod inv, factorials, combinatorics, [in	96	6.25	p4

Problem Name	Problem Code	Status	Submit Count	Reading Time(m)	Thinking Time(m)	Coding Time(m)	Debug Time(m)	Total Time(m)	Problem Level /10	By yourself?	Category	1-2 line Comments about your approach is interesting?	Mostafa Category	Category Code	Level	Quality
AC Averages =>		0	0	0	0	0	0	0	0	0	0	0				
UVA 11327													math, totient	97	3	
SPOJ DCEPCA03													math, totient	97	4.5	
CF101778-GYM-C												Sol	math, totient	97	5	p2
CF1009-D2-D													math, totient, [cases]	97	5	
UVA 11424												Sol	math, totient, sieve, sums, gcd	97	5	
UVA 10820													math, totient	97	5.5	
UVA 11426													math, totient	97	6	p3
UVA 10990													math, totient	97	6	
CF114-D2-F													math, totient	97	6.25	p2
CF100857-GYM-F													math, totient	97	6.25	
TIMUS 1456													math, totient, sieve	97	6.25	
LIVEARCHIVE 2116													math, mobius	98	4	
CF900-D2-D												Sol	math, mobius, inclusion-exclusion or dp_co	98	6	p3
SPOJ SQEBEE												Sol	math, mobius, inclusion-exclusion	98	6.25	p4
CF803-D12-F													math, mobius, inclusion-exclusion	98	6.25	p3
SPOJ SUB_PROB												Sol	string processing, aho_corasick or suffix arr	103	4	
SRM519-D1-500													string processing, aho_corasick	103	5	
SRM557-D2-1000													string processing, aho_corasick, dp or kmp,	103	5	
LIVEARCHIVE 5064												Sol	string processing, aho_corasick	103	6	
TIMUS 1269													string processing, aho_corasick	103	6	
CODECHEF LYRC													string processing, aho_corasick, dp	103	6	
UVA 12244												Sol	string processing, aho_corasick or suffix arr	103	6.25	p2
SPOJ SUBST1													string processing, suffix array, lcp, distinct s	104	3	p3
SPOJ MINMOVE												Sol	string processing, suffix array, lcp, smallest	104	4	p1
SPOJ SUBLEX												Sol	string processing, suffix array	104	4	
UVA 11107													string processing, suffix array, lcp, binary se	104	5	p5
SPOJ LPS												Sol	string processing, suffix array, lcp or rolling	104	5	p3
SPOJ LONGCS												Sol	string processing, suffix array, lcp, LCS k su	104	5	p2
CF113-D1-B													string processing, suffix array	104	5.5	
CF123-D1-D													string processing, suffix array or suffix autom	104	5.5	
LIVEARCHIVE 4477													string processing, suffix array, [solve first U]	104	5.5	
CF149-D2-E													string processing, suffix array or kmp or sufl	104	6	p4
CF129-D2-D													string processing, suffix array, Kth lexograp	104	6	p4
SPOJ PLD													string processing, rolling hash, longest palin	105	4	p1
SPOJ ELCS													string processing, rolling hash	105	4	
CF1003-D3-F													string processing, rolling hash	105	5	p2
CF533-D1-E													string processing, rolling hash	105	5.5	p3
CF101741-gym-K													string processing, rolling hash	105	6	p3
CF101808-gym-B													string processing, rolling hash	105	6	p3
HACKER kniti-and-her													string processing, rolling hash, mo's algorithm	105	6	p3
CF245-D12-H													string processing, rolling hash, dp	105	6	p2
CF1056-D12-E													string processing, rolling hash, math	105	6	p2
CF101864-GYM-J												Sol	string processing, rolling hash, two pointer	105	6	p2
CF727-D2-E													string processing, rolling hash	105	6	
CF985-D12-F												Sol	string processing, rolling hash	105	6.25	p4
CF101627-GYM-D												Sol	string processing, rolling hash, greedy, bina	105	6.25	p3
SPOJ TWIST													bst, treap	106	5	p2
SPOJ ADAAPHID												Sol	bst, treap or segment tree	106	5	p2
SPOJ HEAPULM													bst, treap or cartesian tree or order-statist	106	5	
CF101864-GYM-K												Sol	bst, treap	106	6	p3
SPOJ CERC07S												Sol	bst, treap, segment tree, lazy or splay tree	106	6	p3
kattis.hanoi18.lazylear												Sol	bst, treaps, impl, offline query answering o	106	6	p3
SPOJ GSS6												Sol	bst, treap or splay tree or avl tree, [solve G	106	6	p2
UVA 12003												Discu	bst, treap, impl	106	6	
CF38-D12-G													bst, treap, binary search	106	6.25	p3
SPOJ MEANARR												Sol	bst, treap or bit	106	6.25	p2
SPOJ DQUERY													mo's algorithm or segment tree persistence	107	4	
CODECHEF JITT15													mo's algorithm, bit, [count inversion]	107	5	p1
CF220-D1-B													mo's algorithm	107	5	
HACKER substrings-2													mo's algorithm	107	5	
SPOJ RACETIME												Sol	mo's algorithm or segment tree or sqrt deco	107	5	
SPOJ ZQUERY												Sol	mo's algorithm, [strict time/mem], [!=CODE	107	6.25	
Timus 1167													dp, dp_d&c_opt or dp	112	5	
CF321-D1-E													dp, dp_d&c_opt or dp_knuth, [standard prot	112	6	p2
SPOJ NKLEAVES												Sol	dp, dp_d&c_opt, [standard], [!=UVA 12524, 1	112	6	
HACKER sprint5-minio												Sol	dp, dp_d&c_opt or dp_convex_hull or dp_ki	112	6.25	p4
CF834-D2-D												Sol	dp, dp_d&c_opt, segment tree, [Mido: Wher	112	6.25	p4
TIMUS 1553													graph, hld, lca	122	5	p2
SPOJ QTREE3												Sol	graph, hld or segment tree or bfs	122	5	
SPOJ QTREE													graph, hld, lca, segment tree or splay tree, [	122	5	
SPOJ GRASSPLA												Sol	graph, hld, segment tree	122	5	
PKU 2763												Sol	graph, hld or lca, bit	122	5.5	p2
SPOJ GOT												Sol	graph, hld or lca, segment tree, persistent o	122	6	p2
CF101856-GYM-E												Sol	graph, centroid-decomposition, primes	123	5	p2
CF321-D1-C													graph, centroid-decomposition, [tutorial prot	123	5.25	
CF786-D2-E													graph, centroid-decomposition or dp_trees,	123	5.5	p3
CF101174-GYM-F												Sol	graph, centroid-decomposition or dsu-on-tre	123	5.5	p3
CF914-D12-E													graph, centroid-decomposition, bitmasks or	123	6	p3
CF715-D1-C													graph, centroid-decomposition, dp, math or	123	6	
LIVEARCHIVE 5133												Sol	dp, dp_convex_hull, [needs bst (e.g. set) n	124	6	p4
SPOJ ACQUIRE [7]												Sol	dp, dp_convex_hull, [type 1], [in editorial], [t	124	6	p3
CF319-D1-C												Sol	dp, dp_convex_hull or LiChao, [straightfow	124	6	
CF1083-D1-E												Sol (not intended)	dp, dp_convex_hull or segment tree, LiChai	124	6	
CODECHEF CYCLRA												Sol	dp, dp_convex_hull, dynamic, [https://githu	124	6.25	p3
CF311-D1-B												Sol	dp, dp_convex_hull, [practice problem]	124	6.25	
CF570-D2-D													graph, dsu-on-trees or graph, euler tour, bin	125	4	p1
CF208-D2-E													graph, dsu-on-trees or trees, dfs, binary se	125	4	
HACKER the-grass-ty													graph, dsu-on-trees	125	5	
CF246-D2-E													graph, dsu-on-trees, [standard]	125	5	
SGU 507												Sol	graph, dsu-on-trees, [standard, . (Main a pr	125	5.5	p2
CF1009-D12-F													graph, dsu-on-trees, [standard]	125	5.75	p2
CF291-D2-E												Sol	graph, dsu-on-trees or kmp on tree	125	6.25	p3
CF103-D1-D													sqrt decomposition	126	5	
SPOJ FREQ2												Sol	sqrt decomposition or Mo's algorithm	126	5	
CF797-D12-E													sqrt decomposition, dp	126	5.5	p2
CF342-D2-E													sqrt decomposition, bfs or centroid-decomp	126	5.75	p3
CF13-D12-E													sqrt decomposition	126	6	p3
HACKR competitive-4												Sol	sqrt decomposition, dsu or segment tree [ht	126	6	p3
CF551-D2-E													sqrt decomposition, binary search	126	6	
CODECHEF DOCSDE												Sol	sqrt decomposition, datastructures	126	6.25	p3
UVA 10304												Use knuth	dp, dp_knuth, [standard], [!=UVA 12057]	128	5	p2
UVA 12836												Use knuth	dp, dp_knuth, [standard]	128	5.25	
SPOJ BRKSTRNG												Sol	dp, dp_knuth, [standard]	128	6	p3
CODECHEF CHEFAC												Find O(NK) Sol	dp, dp_knuth, sparse table or dp_d&c	128	6.25	p3
HACKER special-pairs													dp, dp_sos	134	5	p3
CF383-D1-E													dp, dp_sos	134	6	p3
CF165-D2-E													dp, dp_sos, math	134	6	p3

[illegible]

	Weekly Check List
	Training Secrets of Success Video
Reading	
	Read within 3-5 minutes for short text problem. If no, you need to work on your Reading English Skills
	Never suspect later your problem understanding? If happens, you need to improve your comprehension / cases tracing
Thinking	
	Ready and in the challenging mood before start solving.
	Striving against the problem for a reasonable time. If no, you need to change your solving spirit. Be a fighter.
	Found a solution; Do verifications: text cases / extra cases / correctness / time & memory
Coding	
	Sketch in your mind the big picture of the code first. Don't rush for coding
	Code within 10 minutes. If more, you have coding skills problem or your understanding for the approach is not complete
	A lot of copy paste? Something wrong. Need a better code organization
	Needed more than 10 minutes to code medium size codes? Why? Identify the issue and solve it
Debugging	
	Which will be faster to catch the mistake? Printing or Debugger
	Don't know how to use a debugger? Learn this skill
	Needed more than 10 minutes to solve bugs? Something is wrong. Why need all this time? How to solve this issue?
Code is ready!	
	Just submit and see if passed? Wrong. Behave as if you are in the real contest. Are you almost sure it will be AC? If yes, submit
	TRAIN offline as if you are in a real contest. This shortens the gap between training and the real contest
Code Failed :(	
	Are you nervous / frustrated? Yes => Wrong behavior. Take it easy
	Rush to test cases? Yes => Wrong, revise idea, then code, then trace more samples. Try for 15 minutes or more first
Got it AC	
	Read and Understood editorial solutions?
	Checked 1-3 other AC solutions?
	Tried to write a much shorter version of your code?
	Tried to write a faster coder (better complexity)?
Speed	
	How much time do you need in Div2-A/Div2-B? Target (5, 10) minutes for semiseniors, (3, 6) for seniors
	Not that fast? You need regular speed training on easy problems
Weakly contests	
	Do you participate in 1-2 contests per week at least? If no, this is bad. Offline training != Online contests
	You need to train yourself to behave in online contests similar to offline training. This is an important skill.
Sheet stats	
	Recorded them? Yes: read your problem's row. Where do you consume the most of the time? These are your weak skills
	No, I don't record! => How will you know your weak points?!
	Can't record timing as I am mixing thinking with coding? => Wrong behavior. get done with thinking, then move to coding. Don't cycle
Training Time	
	Is it regular and scheduled? Yes => you will have regular improvements.
	Your plan was to train X hours, Did so? If no, why?
	Without regular and continous training, your mind might not improve well
Training with?	
	Yourself only? You may feel bored. If can collaborate with others = longer commitment
Psychological issues	
	Do you keep comparing yourself with others?
	Do you have negative feelings? Like I am stupid...I am hopeless...I will never have a comparable level...?
	Do you think of your image/appearance if failed in online contests so avoid contests?
	Do you use another account with a weird name to train so that people don't know about your progress/failure?
	Do you wish your friends fail in the contest? or get annoyed with their better performance?
	Do you avoid teaching your friends something or give no support to remain better than them?

	Do you feel bored/frustrated as no/weak community in your college?
	Do you keep training day and night without breaks? No socialization at all?
	Do you hate specific topics and avoid them (probability/geometry)?
	'Should I stop' Dilemma? Keep thinking is it worth vs a waste of time?
	If any of the above questions is YES, you probably have a problem and need to find a solution to it.

[illegible]

[illegible]

[1] Google first Goldbach's conjecture

[2] Or shorter:

http://naivered.github.io/2016/07/03/Problem_Solving/UVa/UVa-10168-Summation-of-Four-Primes/

[3] Or shorter:

<https://github.com/ilyesLtifi/Competitive-Programming/blob/master/UVA/UVA%2010325.cpp>

[4] Or shorter:

<https://github.com/ilyesLtifi/Competitive-Programming/blob/master/UVA/UVA%2010325.cpp>

[5] Google first Goldbach's conjecture

[6] Or shorter: http://naivered.github.io/2016/07/03/Problem_Solving/UVa/UVa-10168-Summation-of-Four-Primes/

[7] From a trainee: txt clarification: In this problem FJ can actually REARRANGE the plots and then groups them. * I misunderstood the statement by the word 'successive'